

Automated Aorta Segmentation and Feature Extraction for Medical Analysis

Ruben Filipe Nascimento Abadesso

Dissertação para obtenção do Grau de Mestre em
Engenharia Informática
(2º ciclo de estudos)

Orientador: Prof. Dr. Hugo Pedro Martins Carriço Proença

September, 2026

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Acknowledgments

I would like to express my sincere gratitude to my supervisor, Professor Hugo Proença, for his guidance, expertise, and continuous support throughout the development of this research. His mentorship and valuable feedback were essential to bringing this work to fruition.

This work was supported by a research grant awarded under the R&D agreement established between University of Beira Interior and Çağatay Çivici, facilitated by Empowered Startups. I gratefully acknowledge this support.

Completing this Master's degree proved to be a demanding path marked by numerous challenges, setbacks, and delays. I am deeply thankful to my friends, whose constant motivation, patience, and presence kept me grounded and provided the strength needed to navigate the most difficult phases of this journey.

Finally, to everyone who crossed my path and offered encouragement, whether for a brief moment or along every step of the way, thank you for making this accomplishment possible.

Resumo

As doenças cardiovasculares, particularmente aneurismas e dissecções da aorta, continuam a ser uma das principais causas de mortalidade global. O padrão clínico para diagnóstico depende da angiografia por tomografia computadorizada, contudo, a avaliação de rotina restringe-se frequentemente a medições manuais e unidimensionais, como o diâmetro máximo. Estas métricas simplistas falham na captura das complexas transformações morfológicas tridimensionais associadas à progressão da doença. Além disso, a segmentação manual da aorta é um processo moroso e altamente suscetível à variabilidade entre observadores. Para colmatar estas limitações, esta dissertação apresenta uma infraestrutura computacional totalmente automatizada, concebida para segmentar a anatomia aórtica com precisão e extrair sistematicamente biomarcadores geométricos quantitativos.

A metodologia proposta foi validada num conjunto de dados altamente heterogéneo composto por 56 exames radiológicos. Para superar as diferenças espaciais e de radiodensidade entre aparelhos clínicos, uma rigorosa estratégia de pré-processamento aplicou reamostragem isotrópica e normalização de intensidade. Para a segmentação volumétrica, foi conduzida uma análise comparativa abrangente de cinco arquiteturas de aprendizagem profunda, englobando redes neuronais convolucionais e *vision transformers*, utilizando a biblioteca *MONAI*. Os resultados experimentais estabeleceram a superioridade do modelo *SegResNet*. Beneficiando da sua robusta via de codificação e de ligações residuais, o *SegResNet* alcançou um excepcional coeficiente de similaridade de *Dice* médio de 0.922 e uma distância de Hausdorff no 95º percentil de 36.30 milímetros, mantendo com sucesso a continuidade topológica mesmo em dissecções altamente complexas.

Após a segmentação, as máscaras tridimensionais resultantes são traduzidas em malhas de superfície matemáticas contínuas utilizando o *VMTK*. Auxiliado por uma interface interativa para definir a raiz aórtica, o sistema automatiza a extração de linhas centrais de *Voronoi*. A projeção destes eixos na superfície permite o cálculo de 28 descritores geométricos distintos, incluindo área da secção transversal, excentricidade da forma, curvatura local e tortuosidade cumulativa. Ao converter exames radiológicos em conjuntos de dados tabulares estruturados, esta infraestrutura agiliza a avaliação morfológica e estabelece a base quantitativa necessária para o treino de futuros modelos de aprendizagem computacional para avaliar o risco de rutura.

Palavras-chave

Segmentação da Aorta; Aprendizagem Profunda; SegResNet; Extração de Características Geométricas; Angiografia por Tomografia Computorizada; Análise de Imagem Médica

Resumo alargado

As doenças cardiovasculares, com particular destaque para aquelas que envolvem a remodelação patológica da aorta, como os aneurismas da aorta e as dissecções aórticas, continuam a representar uma das principais causas de mortalidade a nível global. O padrão clínico atual para o diagnóstico, planeamento cirúrgico e monitorização longitudinal destas condições potencialmente fatais depende fortemente da imagiologia por angiografia por tomografia computadorizada. Contudo, na prática clínica de rotina, a avaliação destes exames restringe-se frequentemente a medições manuais e unidimensionais. O diâmetro máximo do vaso é, de forma quase exclusiva, a métrica utilizada para estratificar o risco e ditar a necessidade de intervenção cirúrgica. Embora seja uma métrica clinicamente acessível e amplamente aceite, esta abordagem simplista falha de forma fundamental na captura das complexas transformações morfológicas tridimensionais que estão diretamente associadas à progressão da doença. Alterações críticas, tais como a dilatação luminal assimétrica, a excentricidade transversal irregular e a tortuosidade extrema do vaso, têm um impacto profundo nas forças hemodinâmicas, mas não são refletidas na medição de um único diâmetro.

Além das limitações métricas, a segmentação manual da árvore vascular aórtica, um passo essencial para qualquer análise computacional avançada, é um processo moroso, exaustivo e altamente suscetível à variabilidade entre observadores médicos. Analisar centenas de cortes axiais para isolar a aorta de estruturas anatómicas adjacentes com densidades semelhantes consome recursos hospitalares valiosos. Para colmatar de forma holística estas limitações clínicas e metodológicas, esta dissertação apresenta o desenvolvimento de uma infraestrutura computacional totalmente automatizada. O principal objetivo deste sistema é segmentar a anatomia aórtica com elevada precisão e, subsequentemente, extrair de forma sistemática um vasto conjunto de biomarcadores geométricos quantitativos, convertendo imagens em dados estruturados.

A metodologia proposta foi desenvolvida e validada utilizando um conjunto de dados multicêntrico e altamente heterogêneo, composto por cinquenta e seis exames radiológicos provenientes de diferentes instituições. Esta heterogeneidade apresentou desafios técnicos significativos, nomeadamente devido às discrepâncias nas resoluções espaciais interplanares, nas espessuras dos cortes radiológicos e nas escalas de radiodensidade inerentes a diferentes fabricantes de equipamentos de imagiologia. Para superar estas inconsistências de forma a que os algoritmos processassem a anatomia numa escala física uniforme, foi concebida uma rigorosa estratégia de pré-processamento. Esta etapa aplicou a reamostragem isotrópica para garantir volumes de um milímetro cúbico e uma normalização de intensidade robusta, acompanhada do recorte do espaço de fundo para estabilizar os valores de atenuação radiológica.

Para a fase crucial de segmentação volumétrica, foi feita uma análise comparativa abrangente de cinco arquiteturas de aprendizagem profunda de ponta. Este estudo englobou tanto redes neuronais convolucionais tradicionais como modelos emergentes baseados em *vision*

transformers. Os resultados experimentais estabeleceram, de forma conclusiva, a superioridade do modelo convolucional *SegResNet*. Beneficiando da sua robusta via de codificação assimétrica e de múltiplas ligações residuais, o *SegResNet* provou ser altamente eficiente na extração de características semânticas a partir de um conjunto de dados de volume restrito. O modelo alcançou um excecional coeficiente de similaridade de *Dice* médio de 0.922 e uma distância de Hausdorff no nonagésimo quinto percentil de 36.30 milímetros. De forma notável, o *SegResNet* manteve com sucesso a continuidade topológica da estrutura vascular mesmo perante morfologias patológicas altamente complexas, superando significativamente os modelos baseados em transformadores de visão, que apresentaram dificuldades em generalizar as suas previsões espaciais.

Após a fase de segmentação volumétrica, o sistema transita para a modelação matemática contínua. As máscaras tridimensionais resultantes são traduzidas em malhas de superfície suavizadas, eliminando os artefactos visuais em forma de escada gerados pelas grelhas discretas. Auxiliado por uma interface gráfica interativa desenvolvida para mitigar erros heurísticos na identificação do ponto de origem anatómica (a raiz aórtica), o sistema automatiza integralmente a extração das linhas centrais da estrutura vascular. Esta abstração topológica é alcançada através do cálculo de diagramas de *Voronoi*, assegurando que o caminho gerado rastreia de forma fidedigna o centro matemático do vaso ao longo de toda a sua extensão e através das suas ramificações primárias.

A projeção geométrica destes eixos centrais de volta na malha de superfície permite ao sistema realizar cálculos rigorosos e ortogonais ao percurso do vaso. Através deste mecanismo, a infraestrutura avalia e exporta vinte e oito descritores geométricos e topológicos distintos a intervalos matematicamente regulares. Esta análise abrangente inclui métricas transversais, como a área luminal, a excentricidade e o verdadeiro diâmetro esférico inscrito, e métricas de trajetória espacial, como a curvatura local, a torção e a tortuosidade cumulativa do vaso.

Em suma, ao converter exames radiológicos complexos em conjuntos de dados tabulares altamente estruturados, esta infraestrutura computacional não só agiliza a avaliação morfológica detalhada e isenta de erros humanos, como também estabelece a fundação quantitativa essencial para o futuro da medicina preditiva nesta área. O trabalho futuro focará a utilização desta vasta recolha de variáveis numéricas espaciais para treinar modelos de classificação de aprendizagem automática computacional. Estes futuros modelos preditivos terão a capacidade de distinguir autonomamente bifurcações anatómicas saudáveis de anomalias patológicas locais, melhorando exponencialmente a precisão e a objetividade na avaliação do risco de rutura aórtica na prática clínica diária.

Abstract

Cardiovascular diseases, particularly those involving severe aortic remodeling such as Aortic Aneurysms (AAs) and Aortic Dissections (ADs), remain a leading cause of global mortality. The clinical standard for diagnosis relies on Computed Tomography Angiography (CTA), yet routine assessment is often restricted to manual, unidimensional measurements like maximum diameter. These simplistic metrics fail to capture the complex, Three Dimensional (3D) morphological transformations associated with disease progression. Furthermore, manual segmentation of the Aortic Vessel Tree (AVT) is a labor-intensive process highly susceptible to inter-observer variability. To address these limitations, this dissertation presents a fully automated computational framework designed to precisely segment the aortic anatomy and systematically extract advanced quantitative geometric biomarkers.

The proposed pipeline was validated on a highly heterogeneous multicenter dataset of 56 CTA scans. To overcome spatial and radiodensity discrepancies across scanners, a rigorous preprocessing strategy enforced isotropic resampling and intensity normalization. For volumetric segmentation, a comprehensive comparative analysis of five deep learning architectures, encompassing Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), was conducted utilizing the Medical Open Network for AI (MONAI) framework. Experimental results established the superiority of the *SegResNet* model. Benefiting from its robust contracting encoder and residual connections, *SegResNet* achieved an exceptional average Dice Similarity Coefficient (DSC) of 0.922 and an 95th percentile of the Hausdorff Distance (HD95) of $36.30mm$, successfully maintaining topological continuity even in highly complex dissections.

Following precise voxel-wise segmentation, the resulting 3D masks are translated into continuous mathematical surface meshes using the Vascular Modeling Toolkit (VMTK). Aided by an interactive interface to define the aortic root, the system automates the extraction of *Voronoi* centerlines. Projecting these axes back onto the surface enables the computation of 28 distinct geometric descriptors, including cross-sectional area, shape eccentricity, local curvature, and cumulative tortuosity. By converting discrete radiological scans into structured tabular datasets, this framework streamlines morphological evaluation and establishes the quantitative foundation necessary for training future predictive machine learning models to assess rupture risk.

Keywords

Aorta Segmentation; Deep Learning; SegResNet; Geometric Feature Extraction; Computed Tomography Angiography; Medical Image Analysis

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Acronyms

AA	Aortic Aneurysm
AAA	Abdominal Aortic Aneurysm
AD	Aortic Dissection
ASSD	Average Symmetric Surface Distance
AVT	Aortic Vessel Tree
CNN	Convolutional Neural Network
CT	Computed Tomography
CTA	Computed Tomography Angiography
DSC	Dice Similarity Coefficient
DICOM	Digital Imaging and Communications in Medicine
ESVS	European Society for Vascular Surgery
EVAR	Endovascular Aneurysm Repair
GELU	Gaussian Error Linear Unit
GPU	Graphics Processing Unit
HD95	95th percentile of the Hausdorff Distance
HU	Hounsfield Unit
ILT	Intraluminal Thrombus
IVUS	Intravascular Ultrasound
MICCAI	Medical Image Computing and Computer Assisted Intervention
MONAI	Medical Open Network for AI
MRA	Magnetic Resonance Angiography
MRI	Magnetic Resonance Imaging
NIFTI	Neuroimaging Informatics Technology Initiative
NRRD	Nearly Raw Raster Data
ReLU	Rectified Linear Unit
ROI	Region of Interest

SEG.A.	Segmentation of the Aorta
TAA	Thoracic Aortic Aneurysm
VMTK	Vascular Modeling Toolkit
ViT	Vision Transformer
YOLO	You Only Look Once
2D	Two Dimensional
3D	Three Dimensional

Chapter 1

Introduction

1.1 Chapter Overview

This chapter establishes the foundational context and motivation for the research, highlighting the critical need for automated computational tools in the diagnosis and monitoring of aortic pathologies. It outlines the primary objectives of the dissertation, formally defines the problem statement regarding data heterogeneity and segmentation challenges, and presents the structured work plan and timeline guiding the project's execution. Finally, it provides an outline of the overall dissertation structure, detailing the progression from theoretical background to the final conclusions.

1.2 Context and Motivation

Cardiovascular diseases remain the main cause of mortality globally, presenting a serious challenge for healthcare systems and medical research. Of all the conditions, pathologies affecting the aorta, which is the main systemic artery responsible for delivering oxygenated blood to the entire body, are of particular significance because of their potentially life threatening nature. The structural integrity and morphological state of the aorta is vital in ensuring hemodynamic stability, but this artery is particularly prone to serious pathological remodeling, most notably AAs and ADs.

Currently, the diagnosis of aortic pathologies essentially involves the use of CT scans. In clinical practice, the decision-making process for intervention has often involved the manual measurement of particular biomarkers, specifically the maximum diameter of the vessel. While diameter is a fundamental metric, relying on unidimensional measurements may not fully capture the complex, 3D morphological changes associated with disease progression. For instance, asymmetric dilation or irregular vessel tortuosity can significantly influence hemodynamic stress, yet these factors are difficult to quantify precisely through manual inspection alone.

In addition, the process of manual segmentation and analysis of CT scans is a labor-intensive and time-consuming process. Radiologists are faced with the challenge of dealing with large volumetric datasets to establish the boundaries of the blood vessels in order to isolate the aorta from surrounding anatomical structures. This manual approach is also prone to inter-observer variability, potentially leading to inconsistencies in diagnosis and follow-up monitoring.

With this challenges in mind, there is an increasing need for automated computational frameworks that can efficiently process images with high accuracy. By leveraging advances in deep learning and computer vision, it is possible to develop systems that automatically segment the aorta and extract a large number of geometric descriptors. These automated morphological measurements are not intended to replace standard clinical metrics but rather to complement them, providing clinicians with a more comprehensive quantitative characterization of the vessel’s geometry. Such tools have the potential to enhance the understanding of aortic remodeling and support more informed medical decisions.

1.3 Objectives

The primary objective of this dissertation is to address the current limited availability of comprehensive computational tools for the analysis of cardiovascular pathologies, specifically AA and AD. The research focuses on the development of a fully automated system designed to assist in clinical evaluation and the longitudinal monitoring of disease progression. The core methodology entails the precise segmentation of the aorta and its principal branches, including the brachiocephalic trunk, common carotid arteries, subclavian arteries and common iliac arteries (encompassing both their internal and external segments). Leveraging the resulting 3D segmentation masks, the system generates high-fidelity surface meshes to extract critical geometric features. This process aims to provide clinicians with both advanced visual representations and rigorous numerical analyses of the vessel’s morphology, ultimately delivering an automated software solution that streamlines routine follow-up examinations and facilitates the tracking of pathological remodeling.

To successfully achieve these goals, a structured methodological pipeline is established. Initially, patient CT scans undergo rigorous preprocessing to standardize HU, anatomical orientation, and voxel spacing. This foundational step ensures data consistency and optimal radiodensity distribution for the subsequent deep learning models. Following this, state-of-the-art segmentation architectures are evaluated using the prepared dataset to identify the most effective computational method. Upon selection, the generated 3D volumetric masks are converted into surface meshes, which serve as the basis for extracting quantitative geometric descriptors.

Building upon the extraction of these continuous geometric and topological features, a critical objective of this research is the development of a downstream machine learning prediction framework. Specifically, the structured morphological data will be utilized to train point-wise classification models. These models are designed to autonomously distinguish healthy aortic geometries from pathological anomalies, providing clinicians with automated diagnostic support to identify localized aneurysmal dilations or abnormal structural deformations.

By addressing these objectives, this research aims to advance the field of computer-aided diagnosis by bridging the gap between raw radiological data, geometric quantification, and predictive modeling, thereby enhancing the precision of aortic risk assessment and supporting more informed therapeutic decision-making.

1.4 Problem Statement

However, the development of advanced computerized frameworks for aortic analysis is hindered by data availability and consistency. Due to strict rules regarding patient data privacy, large amounts of publicly accessible data with annotated computed tomography scans are not easily obtained. This limits the amount of training data that can be used for deep learning methods, especially when compared to tasks in the domain of computer vision in general. Furthermore, a high level of diversity can be found among existing data sets. CT scans are found to vary in resolution, slice thickness, as well as noise levels due to differences in acquisition protocols as well as manufacturers.

1.5 Work Planification

- **Task 1 - Dataset Research:** Concurrently with the literature review, research was conducted to identify a suitable dataset containing segmented aortic data. Due to privacy restrictions regarding medical data, the search prioritized public datasets that met the complexity requirements for deep learning applications.
- **Task 2 - Literature Review:** The review focused on medical image segmentation methodologies and feature extraction techniques relevant to the aorta and similar tubular anatomical structures.
- **Task 3 - Implementation of State-of-Art Models:** Methodologies identified during the research phase were implemented. This process encompassed the development of baseline segmentation models and the extraction of geometric features from the aorta.
- **Task 4 - Writing the First Part of the Dissertation:** Following the literature review, the initial chapters of the dissertation were composed. This section covers the review of related work, a description of the selected dataset, and preliminary results derived from the proposed pipeline.
- **Task 5 - Framework Development:** This stage involves the continued development of the computational framework, including a comparison with other state-of-the-art models and the refinement of feature extraction algorithms.
- **Task 6 - Writing the Second Part of the Dissertation:** Upon completion of the framework, the final phase involves documenting the full methodology and presenting a detailed analysis of the experimental results.

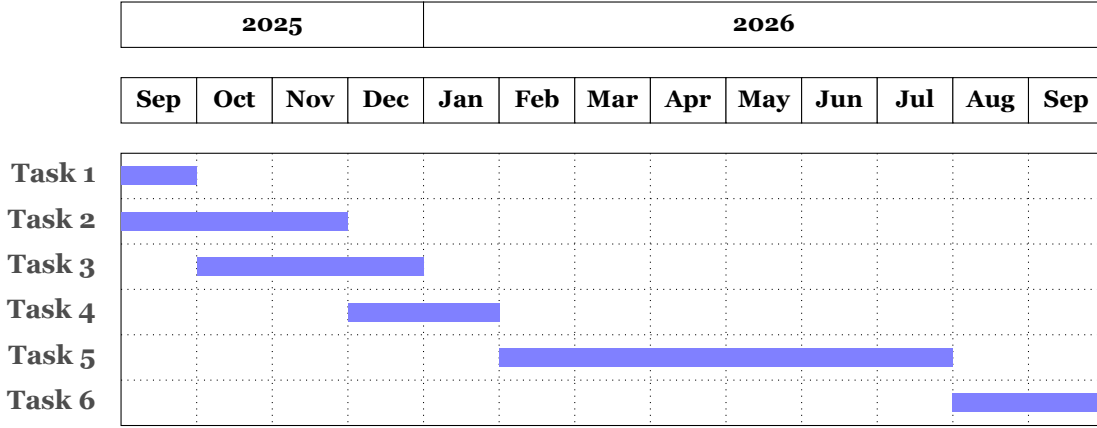


Figure 1.1: Gantt chart illustrating the development timeline and duration of the proposed tasks.

1.6 Dissertation Structure

The present document is structured as follows:

- **Chapter 1 - Introduction:** Introduces the clinical motivation, defines the research objectives regarding automated aortic analysis, and outlines the structural organization of the dissertation.
- **Chapter 2 - Theoretical Background:** Provides the essential medical context on aortic pathologies and geometric biomarkers, alongside the foundational deep learning concepts for medical image segmentation.
- **Chapter 3 - Related Work:** Reviews the evolution of computational methods for aortic segmentation, clinical benchmarking challenges, and established techniques for 3D geometric feature extraction.
- **Chapter 4 - Proposed Pipeline:** Details the automated computational pipeline, from multi-center dataset preprocessing and volumetric segmentation to the topological extraction of morphological biomarkers.
- **Chapter 5 - Experiments and Discussion:** Presents the experimental setup, evaluates the comparative performance of the segmentation architectures, and analyzes the successfully extracted geometric features.
- **Chapter 6 - Conclusions and Future Work:** Summarizes the principal achievements of the research and identifies key areas for future development, including predictive machine learning modeling.

Chapter 2

Theoretical Background

2.1 Chapter Overview

This chapter provides the multidisciplinary foundation required to understand the proposed framework, divided into medical and computational domains. It first explores the medical context, detailing the anatomy of the aorta, relevant cardiovascular pathologies such as aneurysms and dissections, the clinical significance of geometric biomarkers, and the standard imaging modalities used in diagnosis. Subsequently, it introduces the fundamental principles of deep learning for medical image analysis, covering CNNs, encoder-decoder structures, and advanced ViT and hybrid architectures.

2.2 Medical Context

The aorta is the major systemic artery in the human body, from which the entire arterial system arises, and plays the crucial role of maintaining the physiological functionality of the circulatory system. Understanding its anatomy and the pathologies that change its morphology is the key to developing automated segmentation and feature extraction systems. This section provides an overview on aortic anatomy, the cardiovascular diseases that require medical imaging intervention and the morphological biomarkers clinicians use for diagnosis and surgical planing.

2.2.1 Anatomy of the Aorta

The aorta serves as the main vessel of the systemic arterial circulation, originating from the left ventricle of the heart. It is responsible for distributing oxygenated blood to the tissues of the body. The aorta is divided into four distinct segments (Figure 2.1), each with specific geometric characteristics and susceptibility to different diseases:

Ascending Aorta

This segment starts in the base of the left ventricle. It ascends superiorly, slightly forwards and to the right, within the pericardial cavity. The initial portion marks the aortic root where the aortic valve and coronary arteries originate. This part of the aorta is very susceptible to dilatation especially when it comes to connective tissue diseases [14].

Aortic Arch

After the ascending aorta, the aortic arch curves backward and left, forming an arch over the bifurcation of the pulmonary artery and the left main bronchus. This portion of the aorta is geometrically complex because of its high curvature and the presence of supra-aortic arteries, like the brachiocephalic trunk, left common carotid artery and left subclavian artery. The geometric complexity of the aortic arch results in particular hemodynamic shear forces, which contribute to the development of atherosclerosis [15].

Descending Thoracic Aorta

Continuing from the end of the aortic arch until the level of the diaphragm, passing through the posterior mediastinum, this segment is comparatively straight compared with the arch, although it often becomes very tortuous with age and hypertension [16].

Abdominal Aorta

After crossing the diaphragm the aorta enters the abdominal cavity. It continues in front of the lumbar vertebrae, ending by bifurcating into the left and right common iliac arteries, at the level of the fourth lumbar vertebra. This portion of the aorta is the most common site for aneurysms.

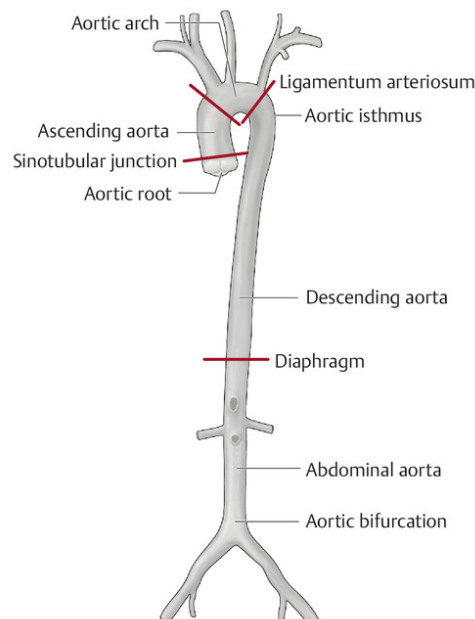


Figure 2.1: Diagrammatic representation of the ascending aorta, aortic arch, descending aorta, abdominal aorta, and the major supra-aortic and abdominal branch vessels [1].

The aortic wall is composed of three layers: the *intima* (inner layer), the *media* (muscular middle layer), and the *adventitia* (outer connective layer) [17] (Figure 2.2). The mechanical integrity of the aorta relies heavily on the elastin and collagen fibers within the *media*. Degeneration or failure of these layers leads to the morphological abnormalities, such as dilation or delamination, that automated systems aim to detect through CT imaging.

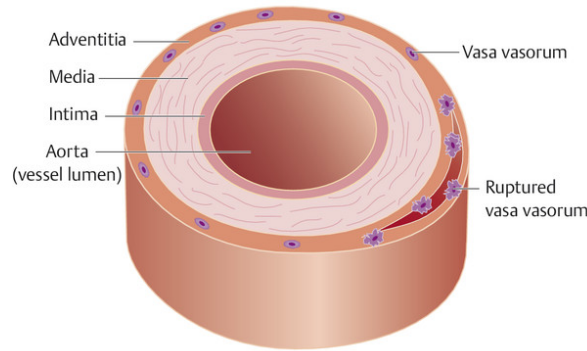


Figure 2.2: Cross-sectional diagram of the aortic wall, illustrating its three primary layers: the intima, the media, and the adventitia. The diagram also highlights the vessel lumen and the vasa vasorum [1].

2.2.2 Pathologies

Cardiovascular diseases involving the aorta can be dangerous and generally occur because of alterations in the geometry of the vessel. The primary focus of computational analysis in this research centers on AAs and ADs.

Abdominal Aortic Aneurysm

An aneurysm is defined as a localized dilatation of an artery having at least a 50% increase in diameter compared to standard size. In the case of the abdominal aorta, a diameter above 3.0cm is clinically recognized as an aneurysm [18]. Because of the absence of symptoms, AAAs (Figure 2.3 & 2.4) have come to be recognized as “silent killers”, as they can grow progressively until rupture occurs. Ruptured AAAs have a mortality rate of over 80% [18], making early detection and surveillance critical.

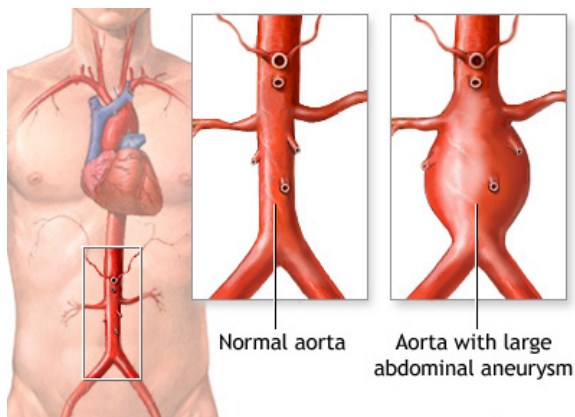


Figure 2.3: Medical illustration contrasting the morphology of a healthy abdominal aorta with one exhibiting a large aneurysmal dilation (AAA) [2].



Figure 2.4: 3D surface mesh reconstruction of a pathological aorta exhibiting an AAA, generated from the automated segmentation of a patient CT scan within the multicenter AVT dataset [3].

Pathophysiologically, AAA is the consequence of the degradation of the extracellular matrix and the loss of elastin [19], resulting in the weakening and dilation of the aorta due to intralu-

minimal pressure. This dilation rarely occurs symmetrically. AAAs often present with complex, non-circular cross-sections and varying degrees of asymmetry. Clinicians currently rely on the maximum diameter as the primary criterion for surgical intervention (typically $> 5.0\text{cm}$ for women and $> 5.5\text{cm}$ for men) [18]. Though it is well understood that diameter alone is an insufficient predictor of rupture risk and more complex geometry, like volume and the curvature of the surface, needs to be evaluated.

Thoracic Aortic Aneurysm

While less common than AAAs, Thoracic Aortic Aneurysms (TAAs) have significant segmentation challenges due to the proximity to the heart and the complex branches of the aortic arch. TAAs are often linked with conditions like Marfan syndrome, Ehlers-Danlos syndrome or atherosclerosis [20]. The geometrical evaluation of TAAs is very important since the complex blood flow patterns within the thoracic cage may contribute to high rates of expansion.

Aortic Dissection

AD is a life-threatening medical emergency caused by a tear in the *intima* layer of the aortic wall (Figure 2.5 & 2.6). This tear allows blood to flow into the *media*, forcing the layers apart. This separation creates an internal membrane, referred to as intimal or dissection flap, that divides the aorta into two separate pathways of blood flow: the true lumen (the original blood channel) and the false lumen (the newly formed channel within the wall). A serious medical complication is possible if blood is separated from flowing inside the aorta. A tear in the aorta can be a consequence of a wide range of factors, such as chronic high blood pressure, connective tissue disorders, atherosclerosis, aging and smoking.

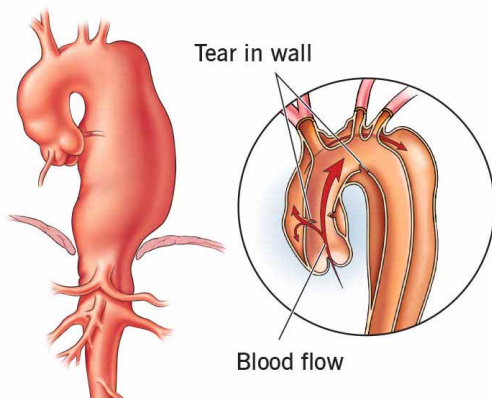


Figure 2.5: Medical diagram illustrating a tear in the internal layer of the aortic wall, which allows blood to flow between the layers and create a false lumen in an AD [4].



Figure 2.6: 3D surface mesh reconstruction of an aorta exhibiting an AD, generated from the automated segmentation of a patient CT scan within the multicenter AVT dataset [3].

From both geometric and segmentation standpoints, ADs are very complex. The separating flap (the intimal flap) can twist spirally along the aorta, creating complex topologies that vary slice by slice in a CT scan. The Stanford classification system divides dissections into Type A (involving the ascending aorta) and Type B (descending aorta) [21]. Type A requires

immediate surgery, while Type B is usually treated either by medication or via endovascular repair in more complicated cases. Automated systems must be capable of distinguishing the exterior boundary of the aorta from the internal structures to accurately calculate the total vessel volume and the compression of the true lumen.

Atherosclerosis and Calcification

Atherosclerosis is the buildup of plaque (fats, cholesterol and calcium) in the arterial walls (Figure 2.7). In CT scans, the calcified plaque show up as high-intensity regions. Although not considered a geometric abnormality of the external boundary of the aorta, calcification does affect the compliance of the vessel and is a major risk factor for the development of aneurysms and dissections. A high concentration of calcification may affect the curvature of the aorta and its eccentricity, thereby disrupting laminar flow and inducing shearing forces against the walls of the vessels.

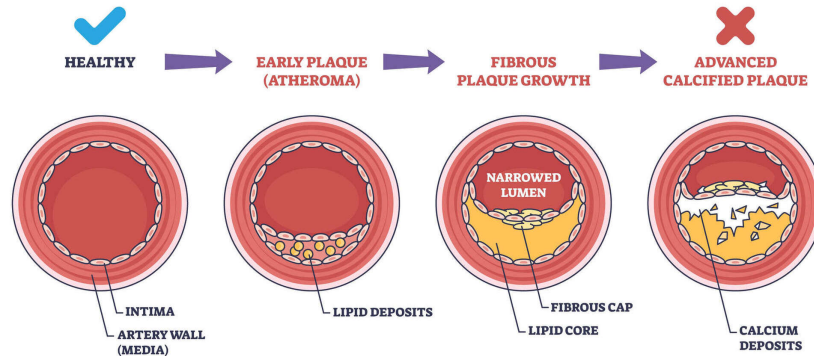


Figure 2.7: Schematic representation of the progression of atherosclerosis, illustrating the transition from a healthy arterial wall to the formation of an advanced calcified plaque and subsequent luminal narrowing [5].

2.2.3 Clinical Relevance of Geometric Features

Although the segmentation of the aorta gives a good graphic visualization of its anatomy, additional quantitative analysis of the morphological measurements may provide complementary information beyond standard clinical metrics. Clinical decision-making has traditionally been dependent on unidimensional measurements, like maximum diameter, despite the aorta's inherently 3D geometry and its susceptibility to pathological remodeling. In this context, a range of geometric features can be computed from segmented volumes to describe shape and spatial organization more comprehensively. Although these features are not yet part of routine clinical practice, they have been explored as potential imaging biomarkers for improving risk stratification in aortic diseases.

Maximum Diameter

The maximum aortic diameter is still the "gold standard" marker that defines the need for surgical intervention. According to guidelines from the European Society for Vascular Surgery (ESVS), elective repair for AAAs is recommended when the diameter exceeds 5.5cm in men and 5.0cm in women, or if the expansion rate exceeds 1cm per year [18].

Despite its widespread use, the maximum diameter has limitations. The maximum diameter calculation assumes that an aneurysm expands evenly and does not take into consideration any possible localized bulges or "blebs", that may be susceptible to rupture despite a smaller overall diameter. In addition, manually measuring aneurysm's diameter on a 2D axial image can lead to errors if the aorta is tortuous and not perpendicular to the scan plane. Automated centerline-based analysis allows for the calculation of the maximum diameter orthogonal to the centerline, thereby providing a much more accurate estimate of aneurysm dimensions.

Aortic Volume

Volumetric analysis provides a global quantification of the aneurysm that diameter measurements often miss. Research suggests that aneurysms can grow in volume without substantial change in maximum diameter [22]. Total vessel volume includes both the blood-filled lumen and the volume of the Intraluminal Thrombus (ILT), a fibrin structure that sits along the inside of the aneurysm wall.

The ratio between the volume of the lumen and the total volume is an important hemodynamic parameter. The measurement of the rate of volumetric growth is increasingly viewed as a more sensitive predictor of rupture risk assessment than measurement of diameter alone [23, 24]. Automated segmentation allows for the precise integration of voxel volumes, removing the inter-observer variability inherent in manual geometric approximations.

Tortuosity

Tortuosity indicates the degree of twisting or turning of the vessel path. It is clinically significant for hemodynamics and surgical planning. When the aorta is highly tortuous, it causes disturbed flow patterns, which may include flow separation and retrograde vortices. These flow disturbances alter wall shear stress, a frictional force that affects endothelial cell function [15]. Areas of low or oscillating wall shear stress are linked with the development of atherosclerotic plaque [25].

In Endovascular Aneurysm Repair (EVAR), a stent-graft is inserted through the femoral arteries to bypass the aneurysm. However, significant tortuosity within the iliac arteries or the aortic neck (the region just below the renal arteries) can make the navigation and deployment of these rigid devices difficult or impossible.

Tortuosity is mathematically quantified using the Distance Metric (ratio of the centerline length to the Euclidean distance between endpoints) or curvature-based metrics.

Curvature and Torsion

While tortuosity describes the vessel globally, curvature and torsion describe the local geometry of the centerline.

Curvature measures the degree to which the vessel deviates from a straight line at a specific

point. High curvature in the aortic arch is a natural physiological feature, but excessive curvature in the abdominal aorta may indicate abnormal tortuosity and be associated with adverse hemodynamic conditions. High curvature regions experience higher mechanical stress on the outer side, making the intima there more susceptible to tears that can lead to dissections.

Torsion, describes how much the aortic centerline twists along its length, reflecting the 3D geometry of the vessel. Higher torsion values indicate a more helical or spiral shape. This is particularly relevant in Type B ADs, where the twisting geometry of the aorta can influence the course and shape of the dissection flap, although other factors such as blood flow and wall stress also play a role.

Eccentricity and Ellipticity

Assuming the aorta is a perfect cylinder is a mathematical simplification that rarely reflects pathological anatomy. Eccentricity, quantifies how much a cross-section deviates from a circle. Ellipticity specifically models the cross-section as an ellipse, defined by the ratio of the major axis to the minor axis.

In AAAs, the vessel often remodels asymmetrically. According to the Law of Laplace, wall tension is proportional to the radius [26]. However, in non-circular vessels, stress concentrates in areas of high local curvature (small radius). Therefore, an aneurysm with a highly elliptical or irregular cross-section may experience peak wall stresses significantly higher than a circular aneurysm of the same equivalent diameter. Quantifying eccentricity and ellipticity provides a more accurate estimation of local wall stress, which can improve rupture risk assessment beyond diameter-based criteria.

Surface Area

The surface area of the aorta can be used as a global geometric descriptor that reflects the enlargement and remodeling of the vessel. In cases such as Marfan syndrome, dilation of the aortic root leads to an increase in surface area as a consequence of increasing radius, accompanying the rise in wall stress driven primarily by diameter, pressure, and wall thickness. In addition to size, the surface area-to-volume ratio provides a quantitative descriptor of aneurysm shape. Compact geometries such as saccular aneurysms approach spherical configurations with lower surface area for a given volume, whereas fusiform, elongated aneurysms exhibit higher ratios. Distinguishing between saccular and fusiform morphologies is clinically important, as saccular aneurysms are often considered higher risk and are treated at smaller sizes, likely due to localized wall weakness and asymmetric stress distributions rather than global size metrics alone.

2.2.4 Imaging Modalities

Medical imaging plays a vital role in the diagnosis, monitoring, and treatment of cardiovascular diseases. While CT is the most widely adopted modality for assessing aortic diseases due to its rapid acquisition and high spatial resolution, Magnetic Resonance Imaging (MRI) provides a powerful alternative with distinct diagnostic advantages. The processing of these medical images requires standardized data formats to enable interoperability between clinical hardware and computational deep learning tools.

Computed Tomography and Angiography

CT imaging measures the attenuation of X-rays as they pass through human body tissues. The attenuation of X-rays varies according to tissue composition. High-density tissues, including bones or calcified tissues, have higher attenuation coefficients compared to low-density tissues such as air or fatty tissues. In a CT image, high-density tissues appear brighter compared to low-density tissues. A CT imaging system consists of a detector that measures various projections of X-rays transmitted from various angles. The detector then uses mathematical algorithms to reconstruct a 3D image of the body. The reconstructed image consists of a stack of axial slices.

However, non-contrast CT imaging may not always provide satisfactory results for vascular imaging owing to the low contrast between the blood pool and the surrounding soft tissues. CTA imaging provides better visualization of the vasculature. In CTA imaging, a contrast agent is injected intravenously to increase the attenuation coefficients of the blood pool. The increase in attenuation coefficients of the blood pool provides better visualization of the vessel lumen. The vessel lumen is clearly differentiated from the vessel wall. In the case of aortic aneurysm and dissection, CTA imaging plays a vital role in imaging the vessel curvature, identifying ILT, and detecting dissection flaps between true and false lumens.

Magnetic Resonance Imaging and Angiography

Unlike CT, MRI does not rely on ionizing radiation; instead, it utilizes strong magnetic fields and radio-frequency pulses to generate detailed cross-sectional images. MRI excels in providing superior soft-tissue contrast, which is highly beneficial for visualizing the aortic wall, identifying intramural hematomas, and assessing surrounding tissues. For targeted vascular assessment, Magnetic Resonance Angiography (MRA) is frequently performed, often utilizing gadolinium-based contrast agents to enhance the blood pool. Despite its excellent soft-tissue resolution and safety profile regarding radiation, MRI is generally less utilized than CT for acute aortic emergencies due to its longer acquisition times, higher operational costs, and patient contraindications, such as claustrophobia or the presence of incompatible metallic implants.

DICOM, NIfTI, and NRRD

For the management and processing of medical images, specific data formats are used. In a hospital setup, the standard format for storing and transferring medical imaging data is the Digital Imaging and Communications in Medicine (DICOM) format. The format combines the pixel values of the medical image with a significant amount of data, including patient information, scanning parameters, and machine settings. Although the format guarantees excellent interoperability between hardware and software in a hospital setup, it is not appropriate for deep learning processing, since the data is generally stored as a series of individual 2D files, one for each axial slice of a volumetric scan.

In order to facilitate the processing of medical image data for deep learning, the data is generally converted from the DICOM format to either the Neuroimaging Informatics Technology Initiative (NIfTI) or Nearly Raw Raster Data (NRRD) format. The NIfTI format is widely accepted by the medical image computing community, since it allows for the consolidation of the entire 3D data set into a single file. The format includes a lightweight header with minimal information, such as spatial information, which is essential for converting voxel coordinates to physical coordinates.

Similarly, the NRRD format was developed to overcome the complexities of DICOM by efficiently representing n-dimensional medical volumes. NRRD is highly flexible, supporting either a single file containing both the header and the raw data (‘.nrrd’) or a detached header file (‘.nhdr’) paired with a separate raw data file. It explicitly defines the spatial orientation, origin, and voxel spacing within its human-readable text header. This explicit spatial definition, combined with its native support in prominent medical image visualization and processing platforms such as 3D Slicer, makes NRRD a highly effective format for 3D volumetric segmentation and geometric feature extraction pipelines. In addition, the conversion from the DICOM format to either NIfTI or NRRD is generally accompanied by the removal of personal health information, ensuring strict data anonymization.

2.3 Deep Learning Fundamentals

The field of automated analysis in medical imaging has undergone a paradigm shift with the development of deep learning, a branch of machine learning that uses multi-layered neural networks. These methods also have the potential for the automated discovery of sophisticated feature representations from raw input data in the form of pixel or voxel intensities in CT images. In the area of cardiovascular imaging analysis, deep learning methods are used for anatomical structure segmentation and the identification of pathologies using the resulted features. This section introduces the fundamental architectures that underlie modern segmentation techniques: CNNs and ViTs [27].

2.3.1 Convolutional Neural Networks

CNNs are the building blocks of many computer vision applications such as image classification, object detection, and semantic segmentation. Inspired by the architecture of the visual cortex, CNNs are designed to process data with a grid structure (Figure 2.8). For instance, unlike fully connected networks where every neuron connects to every neuron in the next layer, CNNs selectively use local and shared weights. This architectural choice makes the network translation-equivariant, meaning that it can pick out a given feature (edges or textures) regardless of where it appears spatially in the input.

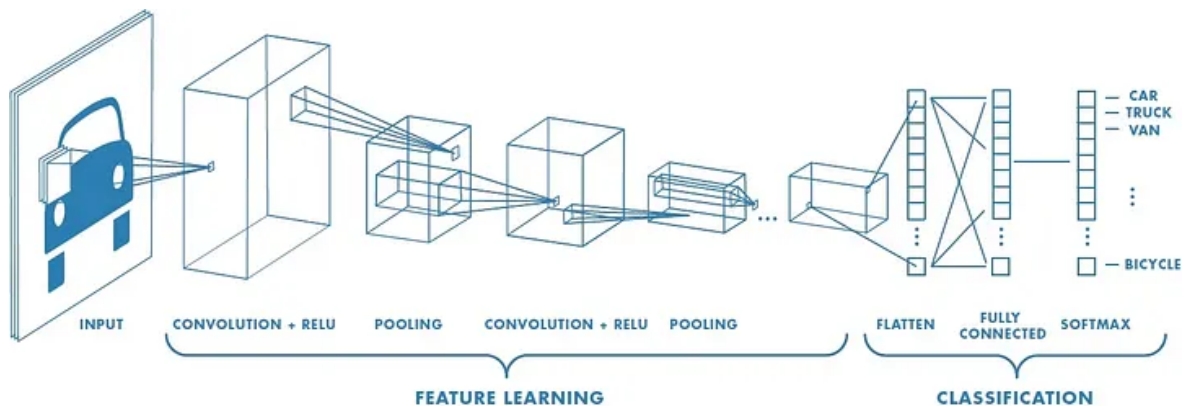


Figure 2.8: Standard CNN architecture for image classification, demonstrating the sequential flow from feature learning, utilizing convolutional, ReLU, and pooling layers, to the final classification stage using fully connected layers and a softmax activation [6].

The basic component of a CNN is the convolutional layer. A convolutional layer works by moving a set of learnable filters or kernels over the input image. At each point, an element-wise multiplication operation is done between the filter values and the input values within the receptive field and then summing the results to get a single value for the output feature map. In the earlier layers of the network, the filters will normally learn to extract information pertaining to basic geometric information including edges, corners, and curves. As the information is propagated deeper into the network, the network attempts to extract more complex information such as the aortic wall or lumen by combining the basic geometric information.

After the convolutional layers, activation functions are introduced to the feature maps. Functions such as the ReLU activation function or its variants, such as the Gaussian Error Linear Unit (GELU) activation function, are used to add non-linearity to the model. Additionally, pooling layers are often used between convolutional layers to successively lower the spatial resolution of the feature map layers. Operations such as max pooling or average pooling are used with the primary goals of increasing efficiency and also to add a sense of translation invariance to feature maps, thereby increasing resistance to minute changes in position of anatomical structures.

3D Convolutions in Medical Imaging

Though conventional CNNs handle 2D images, medical images, specifically CTs angiographies, are inherently 3D volumes. Handling these volumes slice by slice, which involves 2D analysis, can lead to a lack of spatial continuity in the z-axis (longitudinal direction), which can cause inconsistency in segmentation between slices. To handle this, 3D convolutions are used. In 3D CNNs, the filters move through three dimensions: height, width, and depth (Figure 2.9). They learn features by capturing volumetric information.

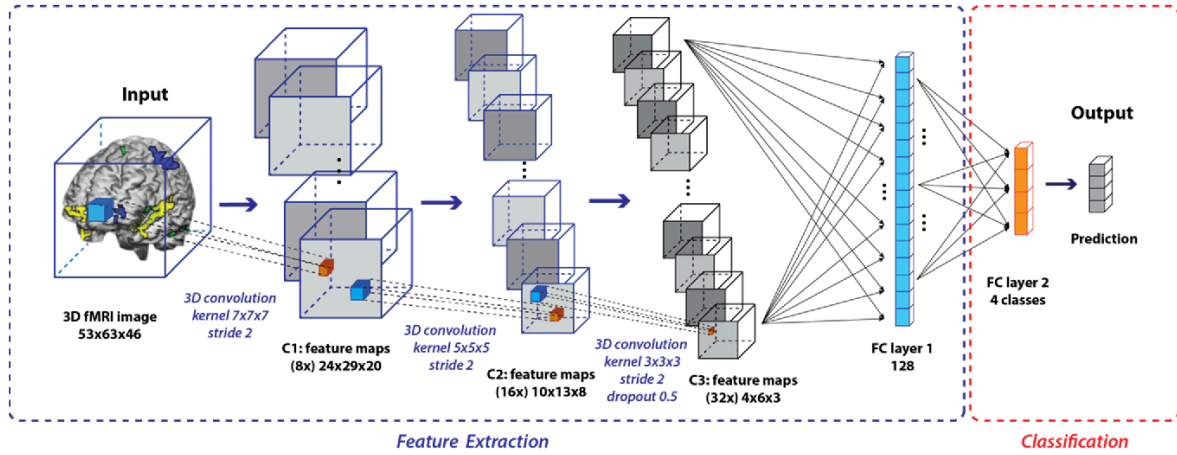


Figure 2.9: Diagram of a 3D CNN architecture. The model processes a 3D input volume through a series of 3D convolutional layers with decreasing spatial dimensions and increasing depth for feature extraction, followed by fully connected layers to generate a final classification prediction [7].

The use of 3D convolution algorithms is specifically relevant in the context of aorta segmentation. This is because the aorta itself is a homogeneous, tubular structure which extends from the thoracic region into the abdominal region in a tortuous fashion in most cases. In this case, the 3D convolution algorithm becomes useful in the sense that it enables the network to utilize information between slices in order to accurately separate the structure from the surrounding tissues of similar radiodensity, such as the *vena cava* or muscle tissue. However, the use of 3D CNN algorithms becomes associated with higher computational costs compared to 2D CNN algorithms, necessitating efficient architectural designs and often requiring the use of patch-based training strategies where the network processes smaller volumetric crops rather than the entire scan simultaneously.

Despite their effectiveness, CNNs have limitations regarding the receptive field. This is due to the localized nature of the convolution process, which means that an individual neuron is only able to process the input from a small region. Although the process of stacking multiple layers helps to improve the receptive field, the ability to process long-distance dependencies effectively is still an impediment for CNNs. This is an important consideration in the case of the aorta, where the connection between the aortic arch and the iliac bifurcation region offers vital context.

2.3.2 Encoder-Decoder

The original application of CNNs was in image classification tasks, where the task is to classify a whole image according to a single label. However, in medical image analysis tasks, a higher level of understanding is also needed. This is because, in the case of extraction of geometric information such as aortic diameter, volume, or tortuosity, semantic segmentation is needed. This is a task where the network has to classify each pixel or voxel in the image according to a label. This helps separate the region of interest from the background. This task is made possible by the use of the encoder-decoder architecture (Figure 2.10).

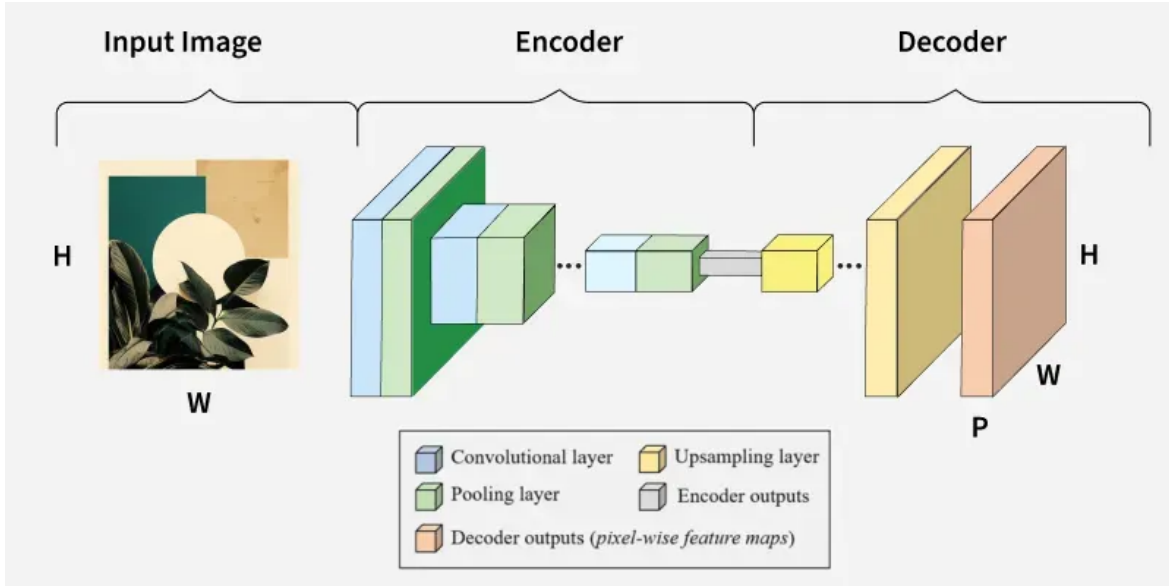


Figure 2.10: Illustration of an Encoder-Decoder network. The encoder processes the input image through convolutional and pooling layers to extract a high-dimensional, low-resolution feature representation in the bottleneck. The decoder then utilizes upsampling layers to recover spatial resolution and output pixel-wise feature maps [8].

An encoder-decoder network consists of two separate pathways that perform complementary functions. The first pathway is the encoder or contracting pathway, and functions like a traditional classification network. It is made up of multiple layers of convolutional layers followed by downsampling layers such as max-pooling or strided convolutions. The main goal of the encoder is to gradually decrease the spatial dimensions of the input feature maps while also increasing the depth of the feature maps. As the information is passed through the pathway, it is forced to ignore the high spatial frequencies while attempting to learn a semantic representation that defines "what" is in the image rather than "where" it is.

The result of the encoder is the low-resolution, high-dimensional feature representation, which is usually denoted as the bottleneck or latent space. While the compressed features in the bottleneck retain the rich semantic information, they lack the spatial resolution to describe the boundaries of the anatomy like the aorta. This issue will be resolved by the addition of the decoder, also called the expanding path. The purpose of the decoder is to enhance the spatial resolution of the data, making it similar in resolution to that of the input, by utilizing transposed convolution or bilinear upsampling layers, followed by convolutional layers.

So, the decoder’s job is to bring back the spatial detail that got lost when the encoder did its downsampling. While progressively upsampling the feature maps, it carries the semantic essence stored in the bottleneck and aligns it with the exact pixel or voxel locations. In the last layer, you normally see a 1×1 convolution (or $1 \times 1 \times 1$ in 3D), followed by a softmax or sigmoid, which gives you a probability map, telling how likely each element is part of the foreground, effectively creating the segmentation mask. The overall design maintains a balance between contextual understanding and precise localization, important for medical tasks, where both the organ’s identity and its actual shape are relevant for further analysis.

2.3.3 U-Shaped Convolutional Networks

While the generic encoder–decoder setup marked a big advance in dense prediction, early versions often produced segmentation masks with blurry boundaries. This is because the downsampling of the encoder alone has lost fine-grained spatial details which upsampling alone in the decoder cannot fully recover. Given that accurately tracing vessel walls in medical imaging is quite necessary to measure features such as eccentricity or surface curvature, edge precision is a must. To address this loss of information, *Ronneberger et al.* [28] proposed the *U-Net*, which has since become the standard for image segmentation.

U-Net

The name *U-Net* comes from its symmetric U-shaped structure (Figure 2.11), which consists of a contracting encoder on the left side and an expanding decoder on the right. The main novelty is represented by the skip connections that link both sides. These connections take feature maps at high-resolution from a given encoder layer and merge them directly with the corresponding upsampled maps in the decoder, preserving fine detail while the network reconstructs the segmentation.

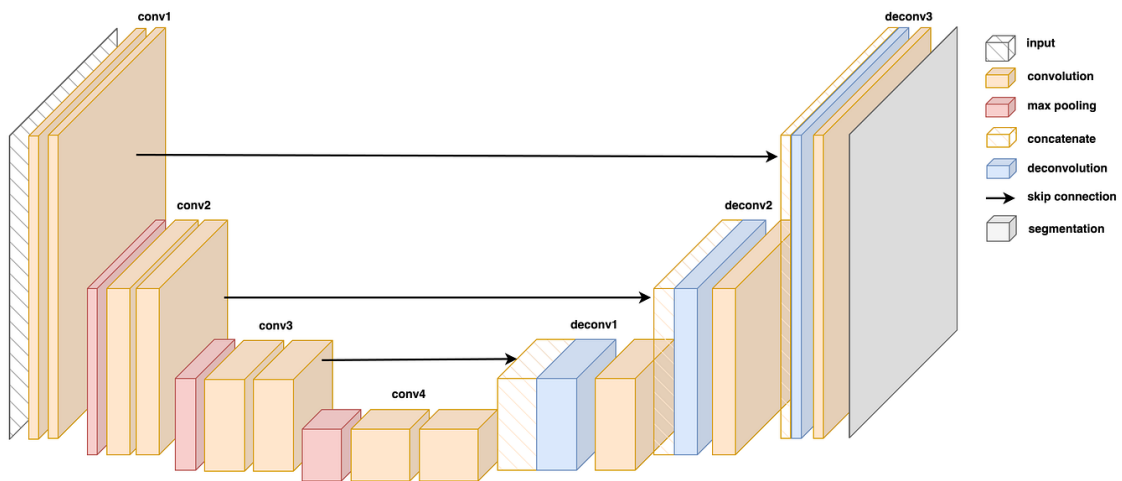


Figure 2.11: Diagram of the U-Net architecture. The model consists of a contracting encoder path on the left for feature extraction and an expanding decoder path on the right. Skip connections link the two sides, concatenating high-resolution encoder feature maps with upsampled decoder maps to preserve fine spatial details during the reconstruction process [9].

Mathematically, let X_l denote the feature map at layer l of the encoder and Y_l the corresponding feature map in the decoder. In a standard encoder-decoder architecture, Y_l depends solely on the upsampled output of the previous decoder layer, such that $Y_l = f(\text{Upsample}(Y_{l-1}))$. In contrast, in the *U-Net* architecture [28], the input to each decoder layer is formed by concatenating the upsampled decoder features with the corresponding encoder features along the channel dimension, yielding $Y_l = f(\text{Concat}(\text{Upsample}(Y_{l-1}), X_l))$, where $\text{Concat}(\cdot)$ denotes channel-wise concatenation and $f(\cdot)$ represents the convolutional operations applied at that layer.

The skip connections allow the direct flow of high-resolution spatial information to bypass the bottleneck. Such a configuration allows the network to leverage the ability to combine the semantic contextual information learned in deeper layers with the textural features learned in the shallower layers. Thus, the network is able to achieve improved precision in the localization of the structures. Such precision is significant in the case of aorta segmentation since the accuracy in the boundary definition directly influences the accuracy in the computation of metrics such as the diameter and area.

3D U-Net

While the inspiration for the original *U-Net* architecture came from 2D biomedical images, structures such as the aorta have a 3D nature. By examining a CT volume on a slice-by-slice basis using a 2D network, there is a possibility of discontinuity along the longitudinal axis of the aorta due to a lack of contextual information regarding the continuity of a 3D volume. This might lead to "staircase" artifacts within a 3D reconstruction, which can affect the computation of centerline-based metrics.

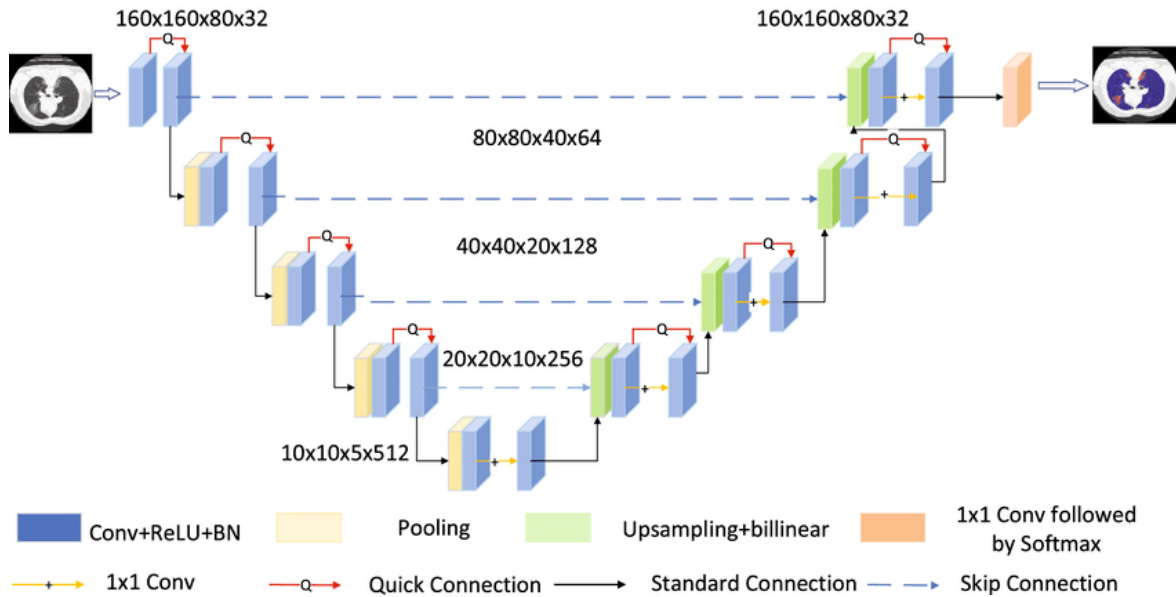


Figure 2.12: Schematic representation of the 3D U-Net architecture. The network adapts the standard U-Net by replacing 2D operations with their 3D counterparts, utilizing 3D convolutions, 3D pooling, and 3D upsampling to process volumetric patches. Skip connections bridge the contracting and expanding pathways to preserve fine-grained spatial details for the final segmentation mask [10].

To overcome this, a 3D variant of *U-Net* was introduced (Figure 2.12). Essentially, in *3D U-Net*, all the 2D operations have been replaced by their 3D counterparts. In this network, 3D convolution operations, 3D max pooling, and 3D upsampling are used to process the patches and obtain information from all three dimensions simultaneously. As a result, the segmentation masks produced will have spatial consistency in the z-axis due to the use of inter-slice information.

However, there are significant computational challenges when moving to 3D architectures. This is because the memory requirements for gradients as well as feature maps grow cubically with the size of the input. Despite the computational costs of 3D networks, they are the only option for medical segmentation problems, where maintaining the topology of the organ as well as its smooth surface, is a requisite step towards further analysis.

DynUNet

Building upon the foundational concepts of the *3D U-Net*, the *DynUNet* architecture represents a highly optimized adaptation inspired by the *nnU-Net* framework. While the original framework dynamically infers blueprint parameters via data fingerprints, the *DynUNet* implementation utilized in this research employs a fixed, high-capacity residual topology to ensure consistent volumetric feature extraction. This adaptation integrates residual blocks into the pathways, mitigating the vanishing gradient problem and allowing for the construction of deeper networks capable of learning more complex anatomical representations.

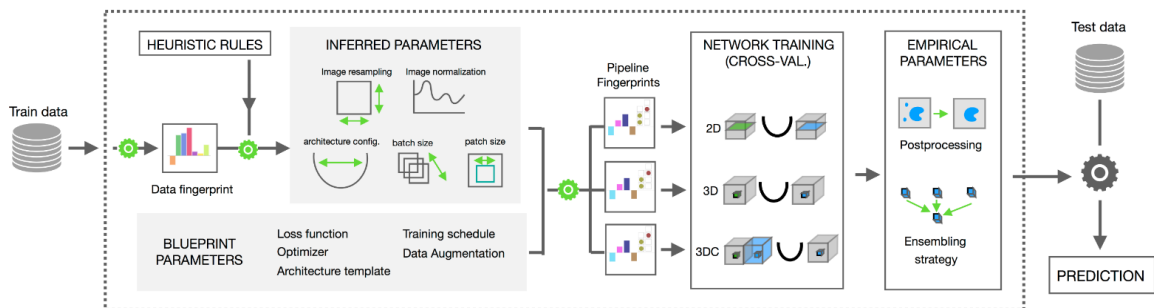


Figure 2.13: Overview of the standard *nnU-Net* framework. While the original framework dynamically infers blueprint parameters via data fingerprints, the *DynUNet* implementation utilized in this research employs a fixed, high-capacity residual topology to ensure consistent volumetric feature extraction [11].

Furthermore, *DynUNet* employs deep supervision during the training phase. By enforcing intermediate feature maps to also contribute to the final segmentation prediction, this technique encourages the network to learn robust, multi-scale representations of the target anatomy, such as the aorta. Utilizing a standardized *DynUNet* topology ensures consistent extraction of volumetric features and facilitates fair performance comparisons across different architectural paradigms without relying on dynamic architectural shifts.

SegResNet

The *SegResNet* architecture introduces a structural asymmetry to the traditional U-shaped paradigm, heavily skewing the computational load toward the contracting encoder pathway. The network features an asymmetric design, dedicating the majority of its depth and residual blocks to the contracting encoder path for robust feature extraction, while utilizing a lightweight decoder to efficiently restore spatial resolution. Recognizing that the extraction of rich, high-level semantic features is generally more complex than the subsequent spatial reconstruction, this design choice enables complex pattern recognition while maintaining overall computational efficiency.

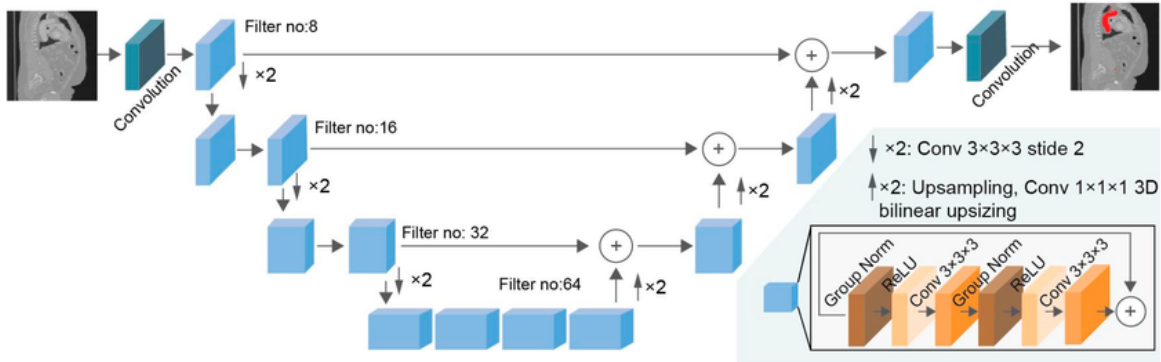


Figure 2.14: Schematic representation of the *SegResNet* architecture. The network features an asymmetric design, dedicating the majority of its depth and residual blocks to the contracting encoder path for robust feature extraction, while utilizing a lightweight decoder to efficiently restore spatial resolution [12].

In contrast to its deep encoder, the decoder in *SegResNet* efficiently restores spatial resolution through upsampling operations, relying on the contextual information passed through skip connections from the encoder. The ubiquitous use of residual connections throughout the contracting blocks allows the network to maintain a smooth gradient flow during training, which is particularly beneficial when processing high-resolution 3D medical volumes. For aortic segmentation, this asymmetric allocation of resources proves highly effective, as the deep encoder captures the global tortuosity and context of the vessel, while the lightweight decoder precisely reconstructs the boundary surfaces required for geometric analysis.

2.3.4 Vision Transformers

In order to overcome the limitations of CNNs in representing long-distance dependencies, computer vision has seen a rise in models based on *Transformers*. *Transformers* are models that were first developed for natural language processing tasks and utilize an attention mechanism to weigh the influence of different parts of the input data on each other. When applied to vision tasks, they are known as ViTs [27].

Standard ViTs, unlike CNNs, receive the image as a sequence of patches. The image is divided into non-overlapping patches, which are further flattened and linearly projected to get

feature embeddings (Figure 2.15). To capture the spatial information, which is present in CNNs but absent in the permutation-invariant architecture of *Transformers*, learnable position embeddings are added to these feature embeddings.

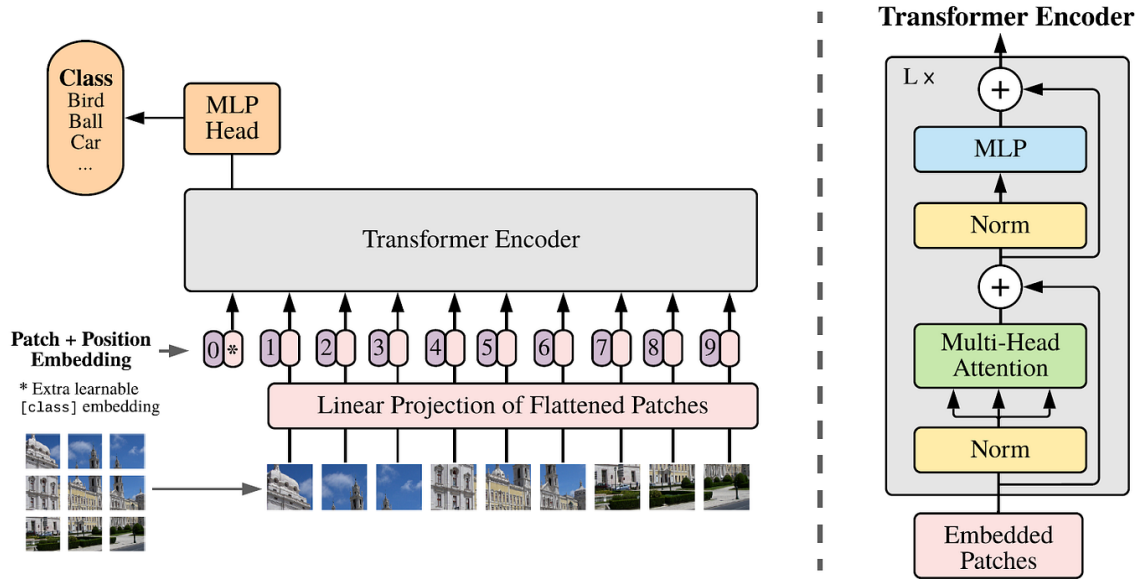


Figure 2.15: Schematic representation of the ViT architecture. The input image is divided into a sequence of non-overlapping patches, which are flattened, linearly projected into feature embeddings, and combined with position embeddings before being processed by the *Transformer* encoder [13].

Self-Attention Mechanism

The central part of the *Transformer* architecture is the multi-head self-attention module. Self-attention allows the network to estimate the relevance of every image patch relative to every other image patch, regardless of the spatial distance between them. This allows the network to capture different anatomical relationships. For example, the network is capable of establishing the relationship between the geometry of the ascending aorta and the geometry of the descending aorta in the same volume scan, facilitating a more coherent segmentation of the entire vessel tree.

In mathematical terms, the self-attention mechanism learns to project the input features into three vectors, namely the *Key*, *Value*, and *Query*. Attention scores are computed as the dot product of the *Key* and *Query* vectors, controlling the extent of the focus placed on different parts of the input. This score is then used as a weight to the *Value* vector, generating an output that considers the global context of the input. While the benefit of the global context is considerable, the complexity of the self-attention mechanism grows quadratically with the number of tokens and becomes an obstacle when scaled to the high-resolution 3D volumes in the medical imaging context.

Hierarchical Transformers and Shifted Windows

In order to counter the computation complexity of the original ViTs [27] but still leverage the benefits of attention-based models, the concept of hierarchical models including *Swin Transformers* [29] was introduced. The *Swin Transformer* model incorporates a hierarchical model similar to that of CNNs but uses window-based self-attention instead of convolution.

In this approach, the input volume is divided into a series of small non-overlapping local windows. The self-attention is calculated only within the local windows, thereby reducing the computational complexity from quadratic to linear with respect to the size of the image. Nonetheless, solely depending on local windows would again impose the receptive field limitation of CNNs. This challenge is overcome by the use of a shifted window partitioning approach, between consecutive layers. This hybrid approach successfully integrates the efficiency of local processing with the global modeling power of *Transformers*. For medical image segmentation tasks, models that incorporate this approach like *SwinUNETR* [30], have achieved state-of-the-art results. They are able to extract the local details needed to define the boundaries of the aortic wall with a global view of the topology of the entire vessel, which is important in analyzing features and geometry.

2.3.5 Hybrid and Transformer-based Architectures

Recent studies have demonstrated that ViTs [27] have strong capabilities in modeling long-range dependencies through self-attention, even though they generally do not possess some of the inductive biases characteristic of Convolutional Neural Networks, namely translation invariance and locality. In the task of segmenting the aorta, both local and global features of the image have been found to be vital. On the one hand, the local features help in accurately specifying the boundaries of the aorta wall, which in turn can help in the precise measurement of the diameter of the aorta. On the other hand, the global features help in the differentiation of the aorta from other tubular structures of similar radiodensity, like the vena cava and the esophagus, and maintaining the continuity of the volume of the aorta in the longitudinal direction. For this reason, hybrid architectures were proposed, incorporating *Transformers* into an existing U-shaped encoder-decoder model.

UNETR

A common hybridization paradigm is to replace the convolutional encoder of the standard *U-Net* with a *Transformer* backbone, giving rise to architectures such as *UNETR* [31] (Figure 2.16). In this configuration, the 3D input volume is split into a sequence of uniformly sized and non-overlapping patches and projected to a linear embedding space. Subsequently, these projected patches are fed through a sequence of *Transformer* layers with global self-attention to extract semantic features of anatomy.

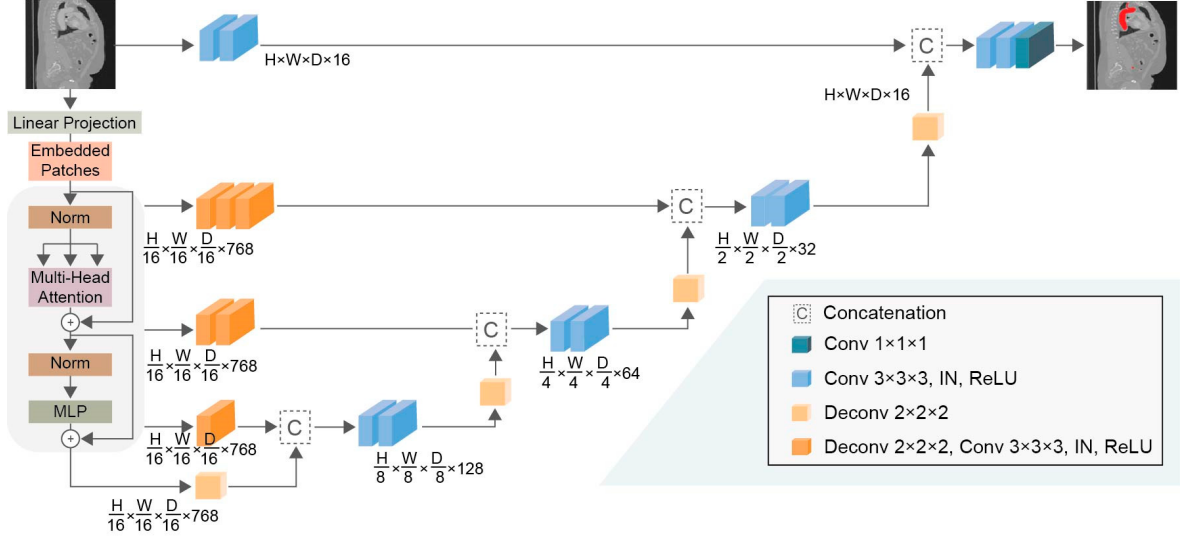


Figure 2.16: Schematic representation of the UNETR architecture. The 3D input volume is split into a sequence of uniformly sized, non-overlapping patches and projected to a linear embedding space. These patches are fed through a sequence of *Transformer* layers to extract global semantic features, which are then rearranged and integrated with a convolutional decoder via skip connections to produce the final segmentation [12].

The features from different *Transformer* encoder layers are rearranged and upsampled to integrate with the convolutional decoder via skip connections. While this configuration is efficient in capturing global anatomical context for the aorta and using the convolutional decoder to improve the resolution of the segmentation mask, it is also computationally expensive for *Transformer* architectures to process high-resolution images and 3D data, due to the quadratic complexity of self-attention mechanisms with respect to the sequence length.

SwinUNETR

To reduce the computation cost of *Transformers* but retain the advantage of the U-shaped structure, the *SwinUNETR* was proposed [30] (Figure 2.17). This architecture is based on the hierarchical structure of the *Swin Transformer* [29] but is applied in the context of medical image segmentation. While regular *Transformers* calculate attention in the whole sequence of the input image, in the case of *SwinUNETR*, self-attention computation is only applied locally within non-overlapping windows. This local computation of attention has reduced the computation cost from quadratic to linear in relation to the size of the images.

Another important feature of the *SwinUNETR* encoder is the proposed Shifted Window mechanism. Although the proposed window-based self-attention calculation reduces computational inefficiency, it also restricts the information within the window boundaries. To overcome this limitation, the *SwinUNETR* architecture utilizes two different partitioning schemes for the windows. In the first layer, the traditional window partitioning scheme is used. In the second layer, the window partitioning scheme is shifted, thus shifting the windows from the previous layer. This shift bridges the boundaries of the previous windows, thus enabling a progressive expansion of the receptive field with the depth of the network.

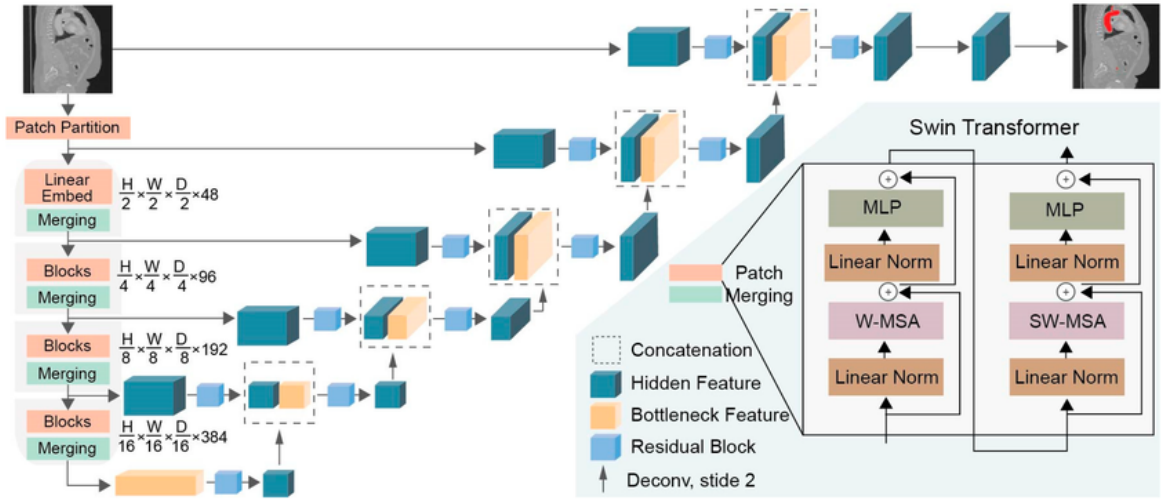


Figure 2.17: Schematic representation of the SwinUNETR architecture. The model utilizes a window-based self-attention mechanism within its hierarchical encoder to process volumetric patches efficiently. Feature maps of varying resolutions are then passed through skip connections to a residual convolutional decoder to accurately reconstruct spatial details for the final segmentation mask [12].

In the case of aortic analysis, the proposed *SwinUNETR* architecture provides a significant advantage. The hierarchical structure of the encoder provides feature maps of varying resolutions, which are connected to a residual convolutional decoder via skip connections. The *Transformer*-based encoder can efficiently model the topological continuity of the aorta in the image volume, thus reducing false negatives in tortuous regions or aneurysmal regions where the geometry deviates from the norm. At the same time, the convolutional decoder and skip connections ensure the preservation of fine spatial details, thus enabling the extraction of smooth surface meshes for the subsequent geometric analysis.

2.4 Summary and Discussion

This chapter established the multidisciplinary foundation necessary for the development of the proposed automated framework. From a clinical perspective, aortic pathologies, particularly aneurysms and dissections, induce complex 3D morphological changes that cannot be fully captured by traditional, unidimensional metrics like maximum diameter. The extraction of advanced geometric biomarkers, such as tortuosity, volume, and local curvature, offers a more comprehensive approach to assessing hemodynamic risk, mapping asymmetrical remodeling, and planning surgical interventions.

However, transitioning from raw radiological data in formats like DICOM or NIfTI to these advanced clinical metrics requires precise volumetric segmentation. As explored in the computational background, deep learning provides the requisite tools for this task. CNNs and U-shaped encoder-decoder networks, such as 3D U-Net and *SegResNet*, excel at capturing the localized spatial details essential for accurate boundary definition. To address the limitations of CNNs in modeling long-range dependencies across complex scans, ViTs and hybrid

architectures like *SwinUNETR* leverage self-attention mechanisms. These attention-based models capture the global anatomical context, which is crucial for maintaining the topological continuity of the AVT and distinguishing the aorta from adjacent structures of similar radiodensity.

In summary, the synergy between clinical anatomy and advanced deep learning establishes the blueprint for the automated diagnostic system. Accurate 3D segmentation serves as the critical intermediate step that translates medical imaging into structured, quantifiable geometric data. The theoretical concepts provide the groundwork for the state-of-the-art methods reviewed in the subsequent chapter, bridging the gap between raw voxel intensities and actionable clinical insights.

Chapter 3

Related Work

3.1 Chapter Overview

This chapter reviews the current state of the art in medical image analysis and geometric feature extraction relevant to cardiovascular modeling. It traces the evolution of computational methods for aortic segmentation from traditional deformable models to modern deep learning architectures, with a specific focus on the methodologies and outcomes of the recent Medical Image Computing and Computer Assisted Intervention (MICCAI) Segmentation of the Aorta (SEG.A.) 2023 challenge and the MONAI framework. Furthermore, it examines existing literature on geometric analysis, detailing established techniques for centerline extraction, cross-sectional morphometry, surface mesh generation, and the calculation of advanced shape descriptors like tortuosity and curvature.

3.2 Medical Image Segmentation

3.2.1 Evolution of Computational Methods for Aortic Analysis

The segmentation of the aorta and the associated vascular tree is a fundamental prerequisite for the assessment of various cardiovascular pathologies, including aneurysms and dissections. In the past, the delineation of the aortic wall from the data provided by the CT scans and MRI scans was primarily reliant on manual or semi-automatic annotations. Although the manual delineation is often used as the ground truth for validation purposes, this is often associated with a number of drawbacks, including a high degree of labor. As a result, the research community has increasingly moved towards fully automatic computational solutions.

Traditional and Deformable Model Approaches

The traditional approaches for aortic segmentation primarily involve model-based approaches and tracking algorithms. The Hough transform was used for detection of circular cross-sections of the aorta from axial slices of images, which are usually used as an initial step toward active contour models as well as level set approaches. These deformable model approaches are usually characterized by energy minimization. The energy function is a combination of the internal forces, which are often responsible for the smoothness of the model, and the external forces, which are often derived from the gradients of the image. These approaches are usually effective for high-contrast CTA images, where the aorta lumen is well defined, but they have failed to deliver satisfactory results in cases of ADs, where the irregular nature of the false lumen, as well as the presence of intimal flaps, may cause significant

topological changes, which are difficult to model using energy minimization approaches [32]. Moreover, non-contrast CT images are characterized by significant complexity, where differentiation of blood pool from surrounding tissues is difficult, which necessitates complex multi-atlas registration approaches for achieving acceptable accuracy of aortic segmentation [33].

Deep Learning and Convolutional Neural Networks

The emergence of deep learning and CNNs has marked a paradigm shift in the analysis of medical images. CNNs eliminate the need for manual feature extraction, as they are able to automatically learn from input data. The *U-Net* architecture [28], characterized by its symmetric encoder-decoder structure and skip connections, has emerged as a standard solution for semantic segmentation tasks. Recent studies have demonstrated the effectiveness of CNNs for the analysis of the heterogeneity of aortic pathologies. For example, *Mavridis et al.* [12] presented a comparative analysis of the effectiveness of the *U-Net* [28], *SegResNet* [34], *UNETR* [31], and *SwinUNETR* [30] architectures for the purpose of segmenting AAAs. The authors concluded that while the *Transformer* architectures have the advantage of considering global context information, the established *U-Net* and the variant of the *SegResNet* network often yielded higher segmentation accuracy and computational efficiency on small datasets.

The application of CNNs is not limited to the lumen segmentation but also to the characterization of different pathologies. In the case of AAA, recent frameworks have used a combination of segmentation and active learning strategies, resulting in high accuracy while reducing manual effort, while also achieving high accuracy in the measurement of morphological parameters such as maximum diameter and volume [35]. deep learning algorithms have also been validated for the automatic detection of aortic dissections, as well as quantification of thoracic aortic enlargement, with a strong correlation with manual measurements carried out by expert radiologists [36].

Transformers and Hybrid Architectures

In spite of the success of CNNs, they also have some inherent limitations in terms of the receptive field, which limits the ability of the model to capture the long-range dependencies in a volumetric scan. To address these limitations, recent studies have explored the application of *Vision Transformers* in medical image segmentation tasks. The *Transformers* use self-attention to weigh the importance of different parts of the data, which allows the model to better comprehend the overall context of the image, rather than the local context of CNNs. However, the application of *Transformers* is not without limitations, as some studies suggest that the application of *Transformers* requires a much larger dataset to achieve better results than CNNs, which may also be computationally more expensive [12]. In the case of intravascular imaging modalities such as Intravascular Ultrasound (IVUS), it is noted that *Transformers* are also being explored in order to effectively differentiate vessel wall boundaries, though bias and interpretability are challenges in these advanced systems [37].

Challenges in Pathological Segmentation

The challenges in the state of the art in this field include the accurate segmentation of the AVT in the presence of severe pathology. In ADs, in which the aortic wall is divided into true and false lumen, is a challenging segmentation problem. The contrast uptake in this case is difficult, and the intimal flap is very thin. According to *Pepe et al.* [32], in spite of several methods being applied to the healthy aorta and simple aneurysms, the accurate segmentation of the entire vessel tree, including the dissection membrane and false lumen, is a challenging problem, and specialized approaches must be applied. The ability of these models in effectively generalizing across multi-center datasets is also a critical area of investigation, and it is noted that domain shifts in these scanners would affect the performance of these models significantly [38].

3.2.2 MICCAI SEG.A. 2023 Challenge

In order to address the challenges associated with anatomical variations and heterogeneity of imaging acquisition protocols, the field of medical image computing has increasingly used standardized benchmarking competitions. The SEG.A. 2023 challenge, which took place in conjunction with the MICCAI conference, aimed at assessing the accuracy of algorithms used in segmentation, modeling, and meshing of the AVT [39]. The main goal of this challenge was to move beyond simple diameter measurements towards fully automated 3D reconstructions, which are not only accurate but also useful for a range of potential applications, including diagnosis, surgical planning, or fluid dynamics simulations.

Dataset Characteristics

The challenge used the multicenter AVT dataset [3], which was designed to represent a heterogeneous population. The dataset consists of 56 CTA scans, with data obtained from three different institutions, including the KiTS19 Grand Challenge [40, 41], the RIDER Lung CT dataset [42], and clinical data from Dongyang Hospital. The dataset includes a range of variations in terms of imaging acquisition parameters, including different manufacturers, resolutions, slice thickness, and reconstruction kernels, as described in table 3.1. The dataset includes a range of healthy as well as pathological cases, including AAAs and ADs. The ground truth segmentations include the entire AVT, including the ascending aorta, aortic arch, descending thoracic aorta, abdominal aorta, and common iliac arteries [3]. This dataset is reviewed in detail in section 4.3.

Challenge Tasks and Evaluation

The MICCAI SEG.A. 2023 challenge had three tasks. The first task was volumetric segmentation of the AVT. Participants had to submit a containerized algorithm that was tested using a hidden test set. The evaluation criteria were the DSC to measure volume overlap, and the HD95 to measure the accuracy of the boundaries. A significant factor was the algorithm’s robustness to different inputs, which was tested by using the original test cases as well as their augmentations [39]. The other tasks were the generation of a surface mesh for visualization

Image Information	KiTS	RIDER	Dongyang
x/y resolution	512 x 512	512 x 512	512 x 666
Axial slices	94/146/1059	260/1008/1140	122/149/251
Slice thickness	0.5/5/5 mm	0.625/0.625/2.5 mm	2/3/3 mm
Pathologies	None	AD, AAA	None
Vessel Tree Volume	55.8/284.5/464.4 ml	176.6/354.1/614.0 ml	126.0/254.9/488.1 ml
Segmentation Times	30/118/30.5 min.	27/422/80 min.	12/35/19.5 min.
Number of Cases	20	18	18

Table 3.1: Image information of the AVT dataset. Number of axial slices, segmentation times, slice thickness and vessel tree volume are given as: min/median/max. [3]

and the generation of a volumetric mesh for simulation. This highlighted the importance of the segmentation being not only voxel-wise accurate but also topologically correct.

Leading Approaches

The results of the challenge highlighted the dominance of deep learning approaches in comparison to the traditional approaches. The winning approach for the first task of the challenge was presented by *Myronenko et al.* [43]. They used an ensemble of 15 models of the *SegResNet* architecture [34], with the models being trained using the 5-fold cross-validation approach. The approach achieved an average DSC score of 0.920 and an HD₉₅ of 6.013mm on the final hidden test set. The most important part of the winning approach was the use of the *Auto3DSeg* pipeline, which is part of the MONAI framework [44], and allows for the automatic configuration of the hyperparameters and extensive data augmentation to deal with the heterogeneity of the source data. The other best-performing methods also used variants of the *U-Net* architecture [28], showing that encoder-decoder architectures remain the state of the art for this specific segmentation problem.

3.2.3 MONAI Framework

The advancement of sophisticated segmentation pipelines in the field of medical imaging has called for the development of specialized computational tools with the ability to efficiently manage high-dimensional volumetric data, as well as domain-specific data formats such as NIfTI and DICOM. MONAI [44] is an increasingly popular, open-source software framework specifically tailored for the implementation of deep learning-based applications in the field of medicine. MONAI is built on the PyTorch library and offers a comprehensive range of modular tools optimized for the implementation of medical image analysis tasks.

One of the primary characteristics of the MONAI framework [44] is the availability of state-of-the-art neural network architectures that are optimized for the implementation of various tasks related to the analysis of medical images. This includes the implementation of well-established CNNs such as the *3D U-Net* [45] and *SegResNet* [34], as well as emerging transformer-based architectures such as *UNETR* [31] and *SwinUNETR* [30]. Recently, several comparative studies have utilized the MONAI library to implement the above architec-

tures for the evaluation of their performance on standardized tasks. For example, *Mavridis et al.* [12] utilized the MONAI framework to evaluate the performance of both convolutional and transformer-based architectures for the segmentation of the abdominal aorta, while *Kim et al.* [35] utilized the MONAI implementation of the *UNETR* [31] and *SwinUNETR* [30] architectures to extract morphological features for aneurysm measurement.

The *Auto3DSeg* pipeline is a feature within the MONAI framework that is a significant advancement towards the automation of the configuration of image segmentation tasks. The pipeline is capable of analyzing the statistical properties of a given dataset and generating heuristic-based hyperparameters that are required for the implementation of the segmentation task. The effectiveness of the *Auto3DSeg* pipeline was demonstrated in the MICCAI SEG.A. 2023 challenge [39], where the winning entry utilized the pipeline to implement an ensemble of *SegResNet* architectures [43]. The framework is capable of autonomously handling critical preprocessing steps like intensity normalization and resampling, thereby enabling the robust handling of the heterogeneity present within the multi-center dataset.

3.3 Geometric Analysis and Feature Extraction

The quantitative evaluation of aortic pathologies goes beyond the primary segmentation of anatomical structures. While semantic segmentation methods provide voxel-wise classification of medical images, clinical decision-making is based on specific morphological and geometric features such as the diameter, volume, tortuosity, and local curvature of vessels. Geometric analysis and feature extraction form the bridge between the volumetric masks obtained from deep learning methods and the quantitative features used by radiologists and surgeons. In recent years, the construction of automated methods for the quantitative evaluation of these features has been reported in the literature. The ultimate aim is to minimize the variability between observers and the time required for the evaluation process. The general workflow for geometric analysis and feature extraction consists of a series of steps, such as 3D surface mesh generation, vessel centerline extraction, cross-sectional measurements, and advanced shape feature calculation.

3.3.1 Centerline Extraction and Skeletonization

The centerline of a vascular structure is considered a fundamental geometric abstraction that follows the central axis of the vessel lumen along its longitudinal axis. It is considered the primary geometric reference for subsequent morphological measurements because standard measurements require strict perpendicularity between the cross-sections and the local vessel axis. In the absence of an accurate centerline, the measurements of the maximum diameter can be significantly overestimated due to oblique cutting planes.

Traditionally, centerline extraction algorithms have been classified into three main types: morphology-based, distance transformation-based, and model evolution-based approaches [46]. Morphology-based approaches, also known as skeletonization, work by iteratively thinning the segmentation mask. Another way of defining the centerline is through the distance mapping approach. This is achieved by calculating the distance from the boundary field within the segmented lumen. The centerline is then determined by the gradient of the distance map. This ensures that the path is always as far as possible from the boundary. In a recent automated algorithm for the characterization of AAAs developed by *Vila Delgado* [47], combined a Gaussian filter for the smoothing of the combined mask of the lumen and the ILT, followed by 3D skeletonization for the determination of the centerline of the vessel. The minimum spanning tree algorithm was used for the determination of the centerline of the vessel. The determined centerline was subsequently smoothed using a spline interpolation function to account for the discrete coordinates of the image voxels.

The 3D *Voronoi* diagram-based approach is an example of an algorithm that calculates paths that have the least total integral of the radius of the maximum inscribed spheres. The *Voronoi* diagram is effective in defining the medial surface of the object. The graph search algorithm is then used to define the shortest path between the endpoints. The *Voronoi* diagram is implemented in the open-source library VMTK, which is used to compute centerlines of branching tubular objects [48].

Recently, deep learning methods have also been investigated for the direct extraction of centerlines from image data or segmentation masks. For instance, *Liu et al.* [46] presented a dual-branch topology-aware *U-Net* (*DTUNet*), which simultaneously predicts the lumen segmentation and vessel centerline. In another study, *Epifanov et al.* [49] presented a multi-task network that used graph layers to directly predict the points of the vessel centerline with real-valued coordinates. In this way, the post-processing steps that are often necessary for voxel-wise skeletonization methods are avoided. In addition, reinforcement learning methods have also been investigated for the detection of anatomical landmarks, such as the aortic root or branch bifurcations, which are then used to guide the extraction of the vessel centerlines [36].

3.3.2 Cross-Sectional and Diameter Assessment

With the smooth centerline now in place, the three-dimensional volume of the aorta can now be analyzed through a series of two-dimensional cross-sections. This is accomplished by discretizing the centerline into points at regular intervals (for instance, every $1mm$ or every $5mm$). The tangent vector is then calculated at each of these points. The cross-section plane is then determined by the orthogonal complement of the tangent vector. The 3D binary mask is resampled along this plane to produce a true cross-section of the vessel.

Automated diameter and area calculations are heavily dependent upon this process. As shown in *Gamechi et al.* [33], automated calculations can determine the cross-sectional average diameter of the aorta at regular intervals relative to specific anatomical landmarks. The maximum diameter can be calculated as the largest distance between contour points. Alternatively, algorithms such as maximal inscribed sphere calculate diameter directly in three-dimensional space by determining the largest sphere that fits inside the vessel boundary at a particular point along the centerline [48].

Besides the regular diameter measurements, cross-sectional analysis can also be used to compute shape descriptors such as circularity, eccentricity, and ellipticity. Circularity is defined as the measure of the similarity of the cross-sectional shape to a perfect circle. A small value of the circularity index can be an indication of an irregular aneurysm shape, the presence of a thrombus, or extensive calcifications, which can be associated with weakened arterial walls.

3.3.3 Surface Mesh Generation and Volumetric Analysis

To perform three-dimensional geometric analysis of the aorta, the segmented masks are usually transformed into surface meshes. The Marching Cubes algorithm is commonly used for isosurface reconstruction from volumetric data. This algorithm results in a mesh comprising vertices and triangular faces. However, due to the anisotropic nature of the CT data, the binary voxel masks may exhibit a staircase effect, necessitating the use of smoothing algorithms to produce realistic representations of the aortic wall.

To perform more advanced meshing for hemodynamic simulations, *Bosnjak et al.* [50] developed a semi-automatic method for creating a block-structured hexahedral mesh for aortic dissections. This method involves creating a topological skeleton which provides a basis for creating a block-structured mesh using transfinite maps.

Once the surface meshes and voxel matrices are defined, the volumetric parameters can be quantified with great accuracy. The total vessel volume, lumen volume, and the volume of thrombus or calcifications can be computed by integrating the voxel sizes in the specific segmented regions.

3.3.4 Tortuosity, Curvature, and Advanced Shape Descriptors

In addition to these basic cross-sectional dimensions, the trajectory and global configuration of the aorta are also crucial determinants of hemodynamic efficiency and surgical intervention. The parameters of tortuosity and curvature are geometric parameters that describe the complexity of the vessel's trajectory. These parameters are especially important in endovascular procedures, where extreme vessel curvature may compromise catheter and stent placement.

Tortuosity is typically calculated using a distance metric that is defined as the ratio of the actual centerline length to the Euclidean distance between vessel endpoints. A value of 1.0 signifies a perfectly linear path, and increasing value signifies increasing degrees of twisting. *Canals et al.* [51] proved the effectiveness of a fully automated pipeline to extract tortuosity feature from the geometry of supra-aortic vessels. The geometric feature was extracted and validated, and relative length and departure angles were compared to manual measurements. The study proved that a fully automated pipeline could be used to obtain objective data to assess catheter access in acute stroke cases, a notion that could be directly applied to aorta access for aneurysm repair.

The local curvature parameter provides a finer assessment of vessel geometry. The mathematical definition of local curvature is given by the reciprocal of the radius of the osculating circle. Local curvature is calculated from derivatives of smoothed centerline coordinates. High curvature is clinically important, especially in areas where there is a high likelihood of abnormal blood flow and high shear stress, contributing to vascular diseases.

Advanced shape descriptors further expand the capabilities of geometric analysis by considering the vessel as a whole 3D object rather than just a tube. Statistical shape modeling is a tool that can be used for the analysis of the global variation of the vessel shapes in a population of patients. *Bruse et al.* [52] developed a non-parametric framework for statistical shape modeling that eliminates the need for manual landmarking. In their study, the authors considered 3D shapes as “currents” (mathematical distributions of shape features). They calculated the mean anatomical template and studied how the individual patient shapes deformed from the mean. This allowed the researchers to identify 3D vessel shape biomarkers, such as specific arch morphologies (for example, “gothic” arches), which correlated with functional parameters such as ejection fraction in patients with repaired aortic coarctation.

Geronzi et al. [53] used a geometric decomposition technique to extract local and global vessel shape features for the prediction of the growth of the AAA. The authors used the ratio of the vessel diameter to the centerline, the ratio of the external to internal curvature lengths, and other descriptors. The results of the study showed that the integration of the global vessel shape modes into regression models can potentially exceed the performance of models using only local measurements, suggesting that the topological configuration of the aneurysm has an independent effect on its progression.

3.4 Summary and Discussion

The literature reviewed in this chapter shows that there is a shift in the field of medical image analysis from traditional, labor-intensive methods to more automated and data-driven methods, enabled by the recent progress in deep learning. The clinical evaluation of aortic

diseases, such as aneurysms and dissections, has traditionally used 2D parameters such as the maximum diameter. However, recent studies have emphasized that such unidimensional parameters are not adequate to capture the complex 3D changes in aortic morphology associated with disease progression and rupture risk.

In the field of semantic segmentation, CNN models, such as the popular *U-Net* and its 3D variants, have provided a solid foundation for anatomical segmentation tasks. Recently, the introduction of *Transformer* modules, such as *SwinUNETR*, has improved the ability of these models to capture complex dependencies, which is a key issue in the segmentation of large, complex structures such as the aorta. The recent results from the MICCAI SEG.A. 2023 challenge have shown that although high overlap can be achieved in the segmentation task, the key issue that persists is the difficulty in maintaining topological accuracy in the segmentation results, especially in the segmentation of pathological structures such as ILT and dissections. The ability to generalize these models to multi-center datasets with diverse imaging protocols is a significant challenge.

Subsequent to segmentation, the extraction of geometric features acts as the interface between voxel data and clinical understanding. The derivation of vessel centerlines is of prime importance, as this enables the computation of accurate dimensions. Although traditional methods based on thinning algorithms are still being used, recent studies indicate that the prediction of vessel centerlines using deep learning can provide improved robustness against artifacts and discontinuities. Moreover, the generation of accurate surface meshes, can aid in advanced volumetric analysis and is essential for the computation of fluid dynamics simulations.

Despite these technological advancements, a significant gap remains in the integration of these individual components into cohesive, fully automated clinical workflows. The current state-of-the-art methods reported in the literature are mostly focused on addressing specific components of the system, either segmentation or centerline prediction, and often require manual intervention for initialization and landmark detection. Furthermore, although advanced shape descriptors are being recognized as promising biomarkers for risk stratification, they are still not being computed in clinical practice due to the lack of standardized and user-friendly computation tools.

The research in this dissertation aims to bridge this gap and create an end-to-end solution using state-of-the-art architectures for segmentation and robust algorithms for geometric analysis. The proposed solution can help in automating the entire clinical analysis pipeline from the initial CT scan data to the generation of morphological reports, which can aid clinicians in decision-making regarding the treatment of aortic pathologies.

Chapter 4

Proposed Pipeline

4.1 Chapter Overview

This chapter details the proposed end-to-end computational framework designed to automate the segmentation of the AVT and the subsequent extraction of geometric biomarkers. It outlines the comprehensive pipeline, beginning with the rigorous preprocessing of the heterogeneous multi-center dataset to standardize spatial resolution and radiodensity. The chapter then describes the volumetric segmentation strategy utilizing deep learning architectures, followed by the topological processing steps required for continuous surface mesh generation, interactive centerline extraction, and the final quantification of morphological features across the system's hardware and software architecture.

4.2 System Overview

The primary objective of the proposed pipeline is to bridge the gap between raw radiological data and actionable clinical insights. To facilitate the clinical evaluation and longitudinal monitoring of aortic disease progression, a highly integrated, semi-automated computational framework was developed. This system is designed to ingest raw CT scans, isolate the AVT, and output a consolidated dataset of quantitative geometric biomarkers.

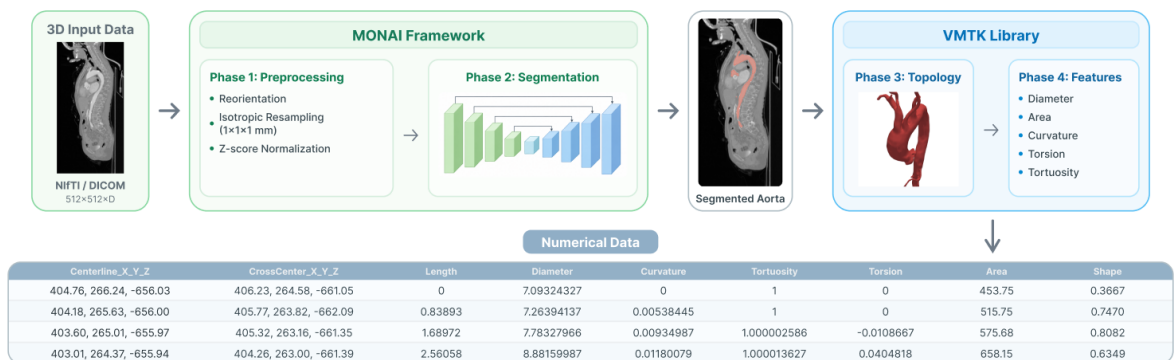


Figure 4.1: Diagram overview of the method's framework.

Rather than treating segmentation and geometric analysis as isolated tasks, the framework connects them through a continuous, sequential pipeline. As illustrated in Figure 4.1, the architecture is divided into four distinct operational stages:

Phase 1: Data Preprocessing

Because the multicenter dataset exhibits severe spatial and radiodensity heterogeneity, the pipeline first standardizes the input. Raw NIfTI volumes are reoriented to a default anatomical axis, isotropically resampled to a $1 \times 1 \times 1\text{mm}$ resolution, and normalized using foreground cropping and Z-score standardization. This ensures the downstream neural network processes anatomy at a consistent physical scale and intensity baseline.

Phase 2: Volumetric Segmentation

The initial analytical phase leverages the MONAI library. The preprocessed volumetric data is passed through a *SegResNet* convolutional architecture. Operating via a sliding window inference mechanism, the model classifies the voxels to delineate the aorta and its principal branches from surrounding tissue. The output is a cleaned, binary segmentation mask (`.seg.nii.gz`) that isolates the largest connected vascular component.

Phase 3: Topological Processing and Centerline Extraction

In the third phase, the discrete voxel mask is converted into a continuous 3D mathematical space using the VMTK. The binary mask is transformed into a smoothed surface mesh via the Marching Cubes algorithm. To ensure accurate topological mapping, the system introduces an interactive graphical interface. The user visually identifies and selects the aortic root on the 3D mesh, establishing a definitive "source point". From this origin, the algorithm automatically extracts the vascular network topology, identifies all distal branch endpoints, and computes the central axes using *Voronoi* diagram.

Phase 4: Quantitative Feature Export

The final stage maps the computed centerlines back to the surface mesh to extract cross-sectional and longitudinal biomarkers. Advanced morphological descriptors, including maximum inscribed sphere radius (diameter), local curvature, torsion, and Frenet frame vectors, are calculated at regular intervals along the centerline path. These multi-dimensional arrays are aggregated and exported as a flattened `.csv` file. Rather than relying on manual clinical evaluation, this structured tabular dataset is explicitly designed to serve as the foundational input for downstream machine learning models, enabling the development of automated predictive algorithms for rupture risk assessment and anomaly detection.

4.3 Dataset Description and Preprocessing

The foundational data for this research is derived from the multicenter AVT collection [3], which was originally established for the MICCAI SEG.A. 2023 Challenge [39]. The cohort comprises 56 CTA scans aggregated from three distinct clinical repositories: the KiTS19 Grand Challenge [40, 41], the RIDER Lung CT dataset [42] and from the Dongyang Hospital.

The dataset captures a broad spectrum of physiological conditions, encompassing healthy

anatomies alongside severe cardiovascular pathologies such as AAAs and ADs. The presence of these structural anomalies introduces significant morphological complexity to the modeling task. The accompanying ground truth annotations offer a continuous, voxel-wise delineation of the entire AVT, extending from the ascending aorta and aortic arch, down through the descending thoracic and abdominal aorta, and terminating at the common iliac arteries. A visual overview of a representative CT volume and its corresponding 3D surface reconstruction is provided in Figure 4.2.

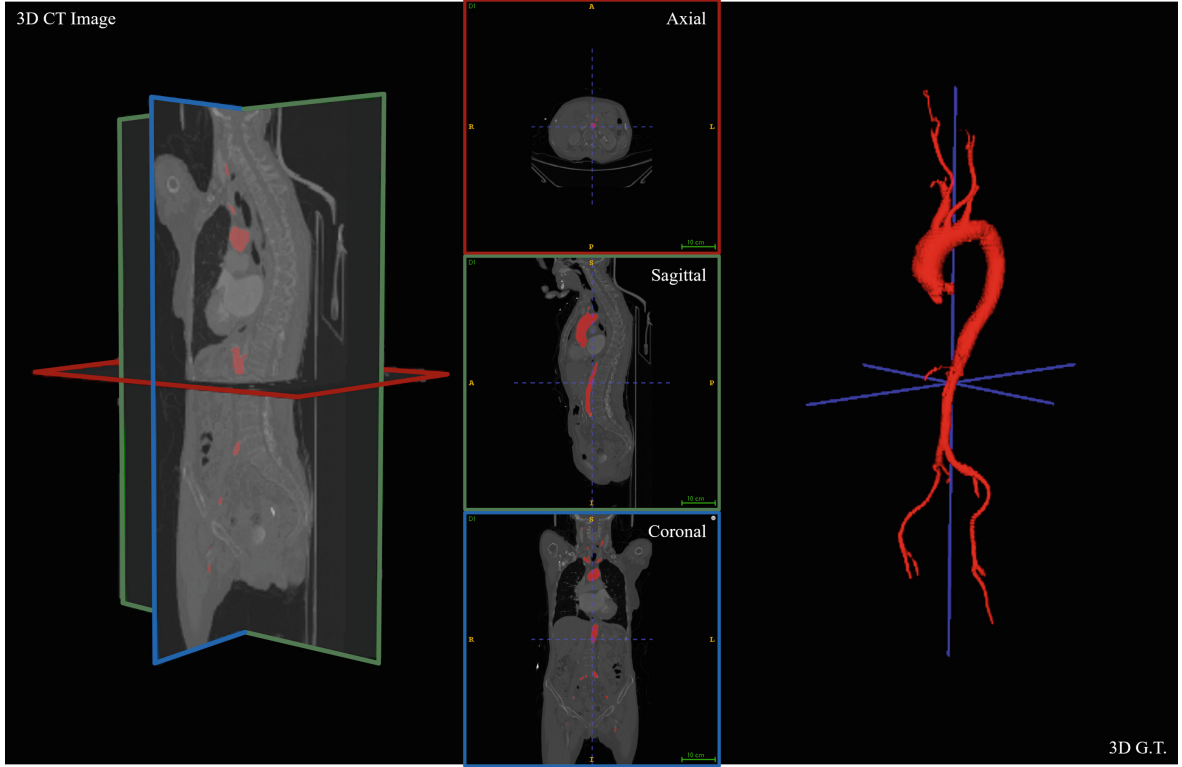


Figure 4.2: Visual representation of a single CT volume from the AVT dataset. The visualization displays orthogonal cross-sectional planes (axial, sagittal, and coronal) with the segmented aorta overlaid in red, alongside the corresponding 3D surface reconstruction of the ground truth segmentation [3].

4.3.1 Spatial Heterogeneity and Isotropic Resampling

A defining characteristic of the AVT dataset [3] is the pronounced heterogeneity of its imaging acquisition protocols. As detailed in table 3.1, the scans originate from various scanner manufacturers, resulting in disparate spatial resolutions, slice thicknesses, and reconstruction kernels across the three clinical sites.

This dimensional inconsistency presents a critical challenge for training deep learning models on multicenter cohorts. In its raw form, the physical space represented by a single voxel varies drastically from one scan to the next. Figure 4.3 demonstrates the high variability in the in-plane spatial resolution (X/Y spacing), which fluctuates between $0.5mm$ and over $1.0mm$ per pixel. The out-of-plane slice thickness (Z spacing) exhibits even more extreme

deviations, as shown in Figure 4.4. While the Dongyang subset maintains a highly standardized thickness of approximately $3.0mm$, the KiTS and RIDER subsets span a wide margin ranging from $0.5mm$ to $5.0mm$.

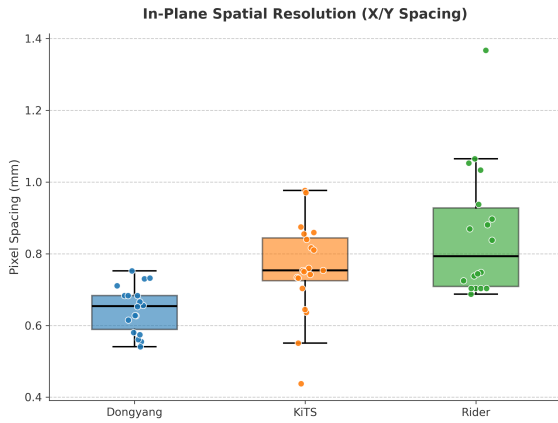


Figure 4.3: In-plane spatial resolution (X/Y pixel spacing) distribution across the three clinical sites.

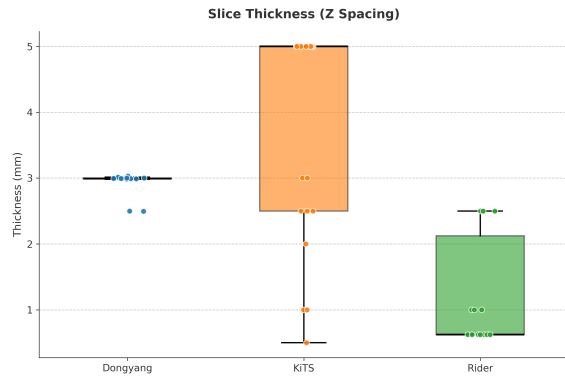


Figure 4.4: Slice thickness (Z-axis spacing) distribution across the dataset.

Consequently, this physical disparity directly dictates the total number of axial slices per scan (Figure 4.5). A high-resolution scan with a $0.5mm$ slice thickness naturally yields a significantly higher slice count to cover the same anatomical region compared to a $3.0mm$ scan. Processing this raw, anisotropic data natively would severely impede the ability of volumetric architectures, such as CNNs or ViTs, to learn scale-invariant geometric features, given that the physical scale of the aorta would shift artificially based on the scanner’s hardware configuration.

To rectify this dimensional mismatch, a mandatory preprocessing step is implemented to isotropically resample all volumes into a standardized $1 \times 1 \times 1mm$ physical coordinate space. Figure 4.6 illustrates the normalized axial slice counts post-resampling. Because each slice now represents exactly $1mm$ of physical depth, the total slice count becomes a direct 1:1 measurement of the scan’s true longitudinal length in millimeters. This transformation guarantees that the network processes all anatomical structures at a uniform and realistic scale.

Additionally, the variance in axial slices points to broader differences in the overall CT field of view. Many full-torso acquisitions include extensive, clinically irrelevant regions such as background air and the scanner bed. To optimize computational efficiency and focus the network’s receptive field strictly on the target anatomy, an automated foreground cropping operation is applied. This step tightly bounds the volumetric array to the patient’s physical contours, stripping away extraneous zero-intensity margins before the data is passed to the network.

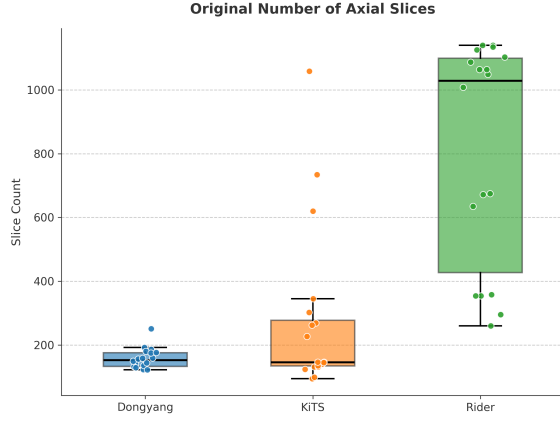


Figure 4.5: Original number of axial slices before spacing normalization.

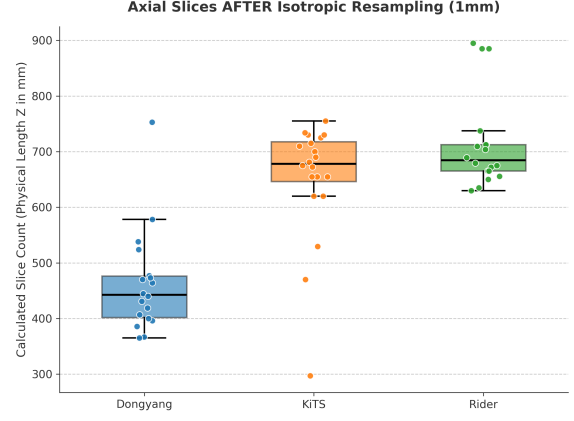


Figure 4.6: Number of axial slices after isotropic resampling to $1 \times 1 \times 1\text{mm}$. The slice count now equals the physical Z-axis length in millimeters.

4.3.2 Radiodensity Shift and Intensity Normalization

Beyond geometric inconsistencies, the multicenter dataset suffers from severe inter-scanner radiodensity variations. While CT radiodensity is universally measured in HUs, discrepancies in institutional protocols, scanner calibrations and image reconstruction padding strategies introduce highly unstable intensity baselines.

Figure 4.7 maps the raw intensity distribution (spanning the 1st to the 99th percentile) for every scan in the dataset. Although the theoretical HU baseline for air is -1000 , the data reveals that only the Dongyang subset consistently adheres to this standard. In contrast, scans from the KiTS [40, 41] dataset frequently shift the air baseline toward 0 or display completely unpredictable minimum values. Consequently, the upper intensity bounds, representing highly dense structures such as cortical bone or metallic artifacts at the 99th percentile, also fluctuate erratically between 500 and 1500 HU. Because deep learning optimization (such as gradient descent) relies heavily on stable, normalized numerical inputs, feeding this chaotic variance directly into a model would severely destabilize the training process.

To achieve numerical stability, the pipeline utilizes a two-stage intensity normalization approach: foreground cropping followed by global Z-score standardization. By shrinking the spatial bounding box to enclose the active foreground prior to normalization, the algorithm discards vast peripheral areas of empty background air. Although the cropped volume remains a rectangular cuboid that still contains some surrounding air and scanner hardware, this significant reduction ensures that the calculated statistical mean and standard deviation are much more heavily influenced by the patient’s tissue. As demonstrated in Figure 4.8, this procedure effectively neutralizes the scanner-induced baseline shifts. The median intensity of every volume is now centered near zero, and the overall intensity bounds are harmonized across the entire multicenter dataset, ensuring robust and consistent feature extraction.

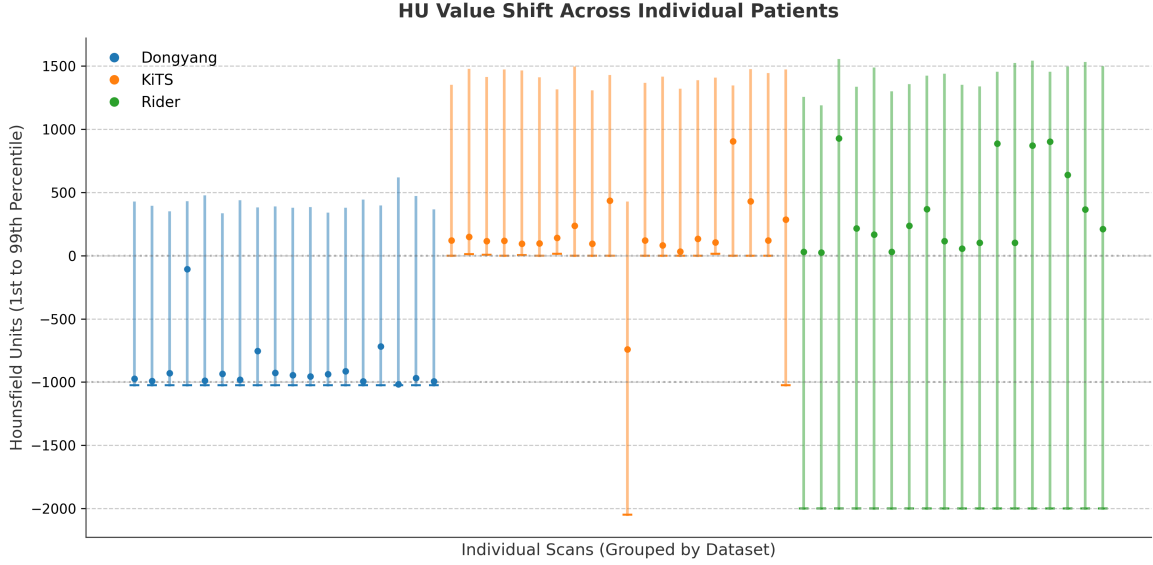


Figure 4.7: Raw HU value shift across individual patients. Each vertical line represents a single scan, spanning from the 1st percentile (approximate air baseline or background values) to the 99th percentile (dense tissue/contrast). The median is indicated by a solid dot. The drastic fluctuations in the air baseline demonstrate the necessity for intensity normalization.

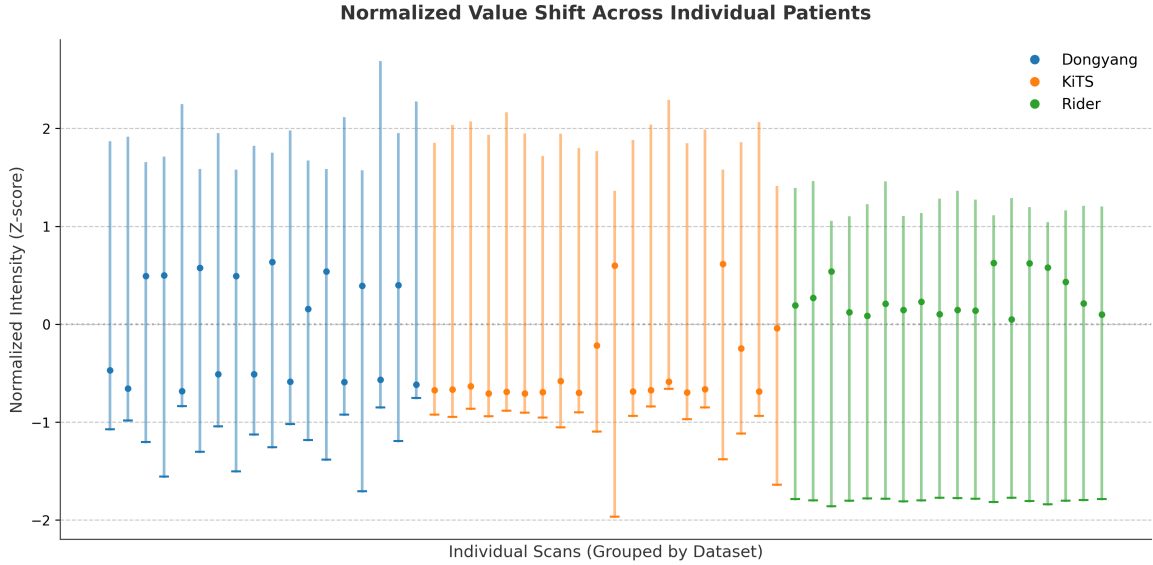


Figure 4.8: Intensity distribution after applying foreground cropping followed by Z-score normalization. The inputs are effectively stabilized around a median of zero, ensuring consistent anatomical representation across all clinical sites regardless of the original scanner baseline.

4.4 Volumetric Segmentation

The first major computational phase of the proposed framework involves the volumetric semantic segmentation of the AVT. Following the rigorous preprocessing detailed in Section 4.3, the normalized 3D volumes are processed to isolate the aorta and its principal branches from the surrounding anatomical structures. This stage is executed utilizing the MONAI framework, specifically leveraging a hybrid ViT-based architecture.

4.4.1 SegResNet Architecture

The core segmentation engine is powered by the *SegResNet* architecture. Unlike transformer-based models that rely on self-attention mechanisms, *SegResNet* leverages a robust 3D convolutional encoder-decoder framework enhanced with residual connections. This design is particularly well-suited for medical image segmentation, as the convolutional layers maintain strong inductive biases for spatial locality and translation invariance, which are essential for accurately delineating the complex and tortuous boundaries of the aorta.

The network progressively downsamples the input volume through a series of residual blocks to capture high-level semantic features, before upsampling the feature maps back to the original resolution to produce dense, voxel-wise predictions. The residual connections mitigate the vanishing gradient problem and ensure the preservation of fine-grained spatial details during the reconstruction process. In this framework, the model is configured with an initial filter size of 48 and a dropout probability of 0.2, processing a single-channel input (the pre-processed CT volume) to yield a two-channel output representing the background and the aortic foreground.

4.4.2 Inference Strategy

Medical CT volumes, even after isotropic resampling, demand significant memory and typically cannot be loaded entirely into Graphics Processing Unit (GPU) VRAM for a single forward pass. To circumvent this hardware limitation, the framework employs a sliding window inference strategy.

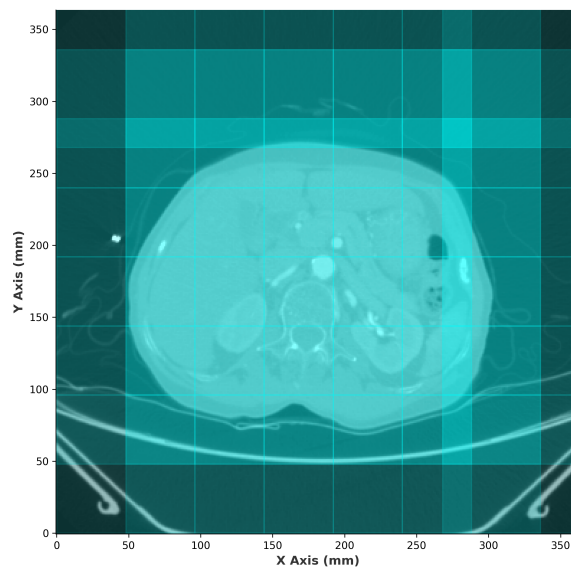


Figure 4.9: Sliding window inference strategy on an axial slice. The translucent grid demarcates $96 \times 96 \times 128$ voxel patches with a 50% overlap (visible as brighter intersecting regions). This intentional redundancy blends predictions via Gaussian weighting to eliminate stitching artifacts in the final 3D reconstruction.

Instead of evaluating the entire volume simultaneously, the model systematically processes smaller, fixed-size volumetric patches, or regions of interest, configured to spatial dimensions of $96 \times 96 \times 128$ voxels. To ensure spatial continuity and prevent "staircase" or stitching artifacts at the boundaries of these patches, the sliding windows are configured with a 50% overlap (Figure 4.9). The predictions from these overlapping regions are subsequently aggregated and blended using Gaussian weighting to reconstruct a full-sized, continuous probability map of the patient's anatomy. For optimal computational throughput, these patches are processed in batches of 8 during inference.

4.4.3 Post-Processing

The raw output generated by the *SegResNet* model consists of uncalibrated spatial logits. A series of automated post-processing transformations is sequentially applied to refine these predictions into a mathematically viable 3D shape.

First, a softmax activation function converts the raw logits into normalized class probabilities. For a given voxel with a logit vector z , the probability $\hat{p}_c$ of belonging to class c is computed as:

$$\hat{p}_c = \frac{e^{z_c}}{\sum_{j=1}^C e^{z_j}}$$

where C represents the total number of classes ($C = 2$ for the background and the aortic foreground). A discrete argmax operation is then executed to assign each voxel to the most probable class, yielding the raw binary mask $\hat{y}$:

$$\hat{y} = \arg \max_{c \in \{0,1\}} \hat{p}_c$$

However, deep learning segmentations are occasionally susceptible to isolated false positives, such as misclassifying an adjacent segment of the vena cava or high-density bone structures.

Because the true AVT is anatomically a single, contiguous tubular structure, a targeted morphological post-processing step is enforced. The system automatically identifies and retains only the largest connected component within the foreground mask, strictly discarding any disjointed, floating artifacts, or noise. Finally, spatial inversion transforms map the refined mask back to its original physical space, and the output is saved as a `.seg.nii.gz` file, serving as the foundational input for the geometric feature extraction phase.

4.5 Geometric Analysis and Feature Extraction

While volumetric segmentation successfully isolates the aorta, a discrete voxel grid is fundamentally inadequate for precise hemodynamic simulation or advanced morphological quantification. The second major phase of the proposed framework translates the binary voxel mask into a continuous mathematical space to compute clinical biomarkers. This transition and subsequent geometric analysis are orchestrated using the VMTK library.

4.5.1 Surface Mesh Generation

The initial step in this phase is the reconstruction of a 3D surface from the binary volumetric mask. The framework employs the Marching Cubes algorithm to generate a highly detailed polygonal mesh representing the aortic wall boundary.

Because the input data originates from discrete voxel grids, the raw generated mesh inherently suffers from anisotropic "staircase" artifacts (Figure 4.10), particularly in regions where the vessel curves obliquely relative to the scanning planes. To mitigate this, a surface smoothing operation is applied. This process carefully relaxes the mesh vertices to produce a continuous, anatomically realistic surface without inducing volume shrinkage or eroding the underlying topological integrity of the vessel (Figure 4.11).

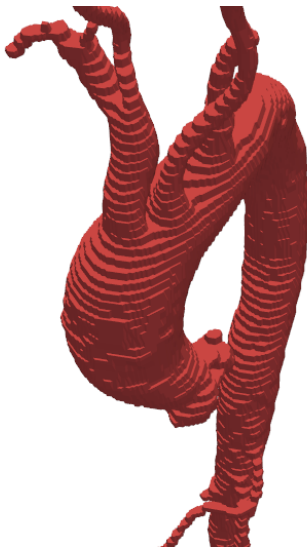


Figure 4.10: Raw 3D surface mesh generated via the Marching Cubes algorithm. The reconstruction exhibits pronounced anisotropic "staircase" artifacts derived from the discrete voxel grid boundaries.

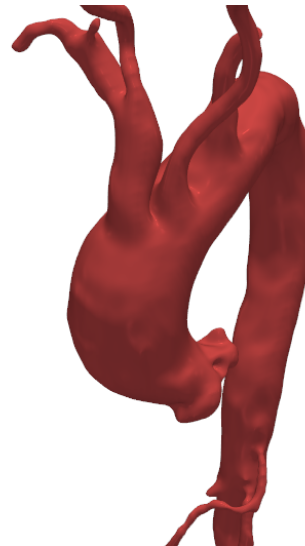


Figure 4.11: Refined surface mesh after applying volume-preserving Taubin smoothing. The vertices are relaxed to produce a continuous, anatomically realistic topology without eroding the underlying vessel structure.

4.5.2 Interactive Source Selection

A critical step in vascular centerline extraction is identifying the origin (source point) and the terminal ends (target points) of the vessel tree. In the proposed framework, the initial detection of all structural extremities is performed automatically. Utilizing network extraction algorithms provided by the VMTK library, the system generates a topological graph of the segmented mesh and automatically extracts the spatial coordinates of every terminal leaf node. This comprehensively captures the distal endpoints of the common iliac arteries, as well as the superior endpoints of the supra-aortic branches, namely the brachiocephalic trunk, common carotid arteries, and subclavian arteries.

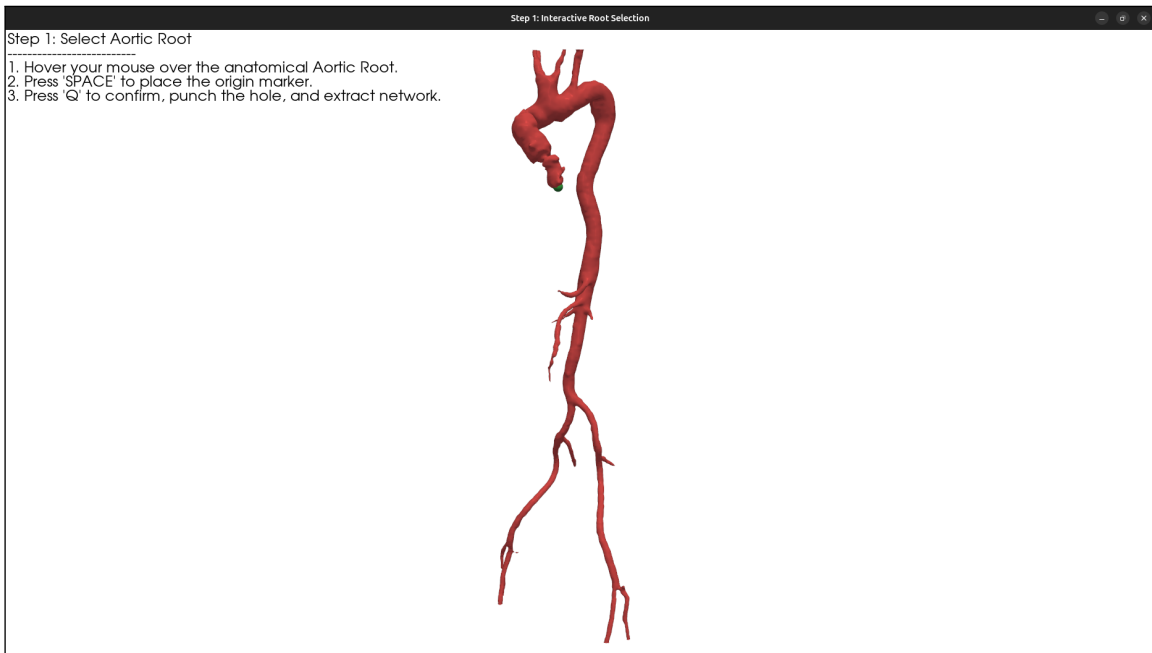


Figure 4.12: Interactive graphical interface for topological mapping and source point selection. The framework renders the smoothed 3D surface and allows the operator to manually define the aortic root via a spatial ray-cast (green sphere). The system then computes the network topology and displays all automatically detected distal endpoints (yellow spheres). This verification stage enables the user to review the network and interactively delete any erroneous anatomical branches before initiating the mathematical centerline extraction.

While the system successfully locates all endpoints, automatically isolating the aortic root (the true origin) from this collection of points presents a significant challenge. A common spatial heuristic is to assume the aortic root corresponds to the endpoint with the highest Z-coordinate. However, this assumption inherently fails in complete AVT segmentations. Because the supra-aortic branches physically extend superiorly toward the head, their terminal endpoints are situated considerably higher along the Z-axis than the aortic root itself. Consequently, simple coordinate-based heuristics will incorrectly misidentify a supra-aortic branch as the vessel origin.

To guarantee accurate topological mapping and overcome this heuristic limitation, the pipeline incorporates a strategic human-in-the-loop interactive interface built with *PyVista* (Figure 4.12). Upon mesh generation, a 3D visualization window is temporarily rendered, prompting

the user to visually locate the true anatomical origin of the vascular tree (the aortic root) and interactively place a definitive marker using a spatial ray-casting tool. Rather than using this manual coordinate directly to draw the path, the algorithm calculates the Euclidean distance between the user-defined click and all the automatically extracted VMTK endpoints. The system designates the single closest automatically extracted endpoint to the user's marker as the absolute source point. The remaining unselected endpoints are subsequently classified as target points, providing the exact mathematical references required for the centerline computation that follows.

4.5.3 Centerline Computation

With the source and target points definitively established, the framework proceeds to extract the central topological axis of the vessel tree. The centerline extraction is driven by the computation of *Voronoi* diagrams inside the bounded surface mesh. The algorithm calculates the paths from the assigned source point to all target points, tracing the trajectory of the maximum inscribed spheres through the vascular lumen. This approach guarantees that the resulting centerlines are robust to local surface irregularities and consistently track the true mathematical center of the vessel, successfully branching at bifurcations to map the entire segmented tree.



Figure 4.13: Smoothed 3D surface mesh of the segmented aorta, reconstructed using volume-preserving Taubin smoothing to ensure anatomical fidelity.



Figure 4.14: Corresponding topological centerline network extracted from the surface mesh, defining the continuous central mathematical axes of the vascular structure.

4.5.4 Morphological Feature Extraction

The final computational step projects the computed centerlines back onto the 3D surface to extract a comprehensive suite of quantitative clinical descriptors. These biomarkers are

calculated at regular intervals along the longitudinal axis of the vessel and are broadly categorized into two groups:

- **Cross-Sectional Metrics:** At each centerline point, the maximum inscribed sphere radius is recorded, providing a continuous measurement of the effective vessel diameter (as visualized in Figure 5.6). This is crucial for identifying localized aneurysmal dilations or stenotic narrowing.
- **3D Path Metrics:** The geometric behavior of the vessel path is quantified through advanced calculus of curves. The pipeline computes local Curvature (Figure 5.9), Torsion (Figure 5.10), and Tortuosity (Figure 5.11), alongside the orthogonal *Frenet* frames (Tangent, Normal, and Binormal vectors). These metrics offer deep insights into the pathological remodeling and bending forces exerted on the aortic wall.

Upon completion, these multi-dimensional arrays are aggregated and exported. The visual representations are saved as `.vtp` geometry files for rendering, while the numerical data is flattened and saved as a structured `.csv` file. This comprehensive feature set is highly optimized for ingestion by machine learning algorithms, facilitating the training of point-wise classification models capable of autonomously evaluating rupture risk and tracking longitudinal disease progression.

4.6 System Architecture

The development, training, and evaluation of the proposed computational framework were conducted utilizing a hybrid hardware infrastructure, combining local development environments with high-performance computing clusters provided by University of Beira Interior.

4.6.1 Software Stack

The automated pipeline is entirely implemented in *Python 3.11*, deployed on an *Ubuntu 24.04 LTS* operating system. The volumetric segmentation phase relies on the *PyTorch* deep learning framework, used in conjunction with the MONAI library, which provides specialized utilities for handling medical imaging transforms and sliding window inference.

For the geometric analysis phase, the framework integrates the VMTK library to manage complex spatial computations, such as *Voronoi* diagram generation and *Frenet* frame extraction. Additionally, *PyVista* is employed to power the interactive 3D graphical interface, enabling real-time surface mesh rendering and the human-in-the-loop ray-casting required for accurate source point selection.

4.6.2 Hardware Infrastructure

The computational pipeline demands varying levels of hardware acceleration depending on the execution phase. Due to the significant memory footprint of processing 3D medical volumes, the highly demanding deep learning training, validation, and full volumetric inference phases were exclusively executed on two dedicated high-performance enterprise server nodes:

- **Server Configuration A:** Equipped with dual *Intel Xeon Silver 4310* processors (2.10 GHz), 128 GB of system memory, and accelerated by 7 *NVIDIA RTX A40* GPUs (48 GB VRAM).
- **Server Configuration B:** Equipped with an *Intel Xeon Silver 4216* processor (2.10 GHz), 128 GB of system memory, and accelerated by 6 *NVIDIA RTX A6000* GPUs (48 GB VRAM).

Conversely, iterative framework development, surface mesh generation, and the interactive centerline extraction were deployed on a local machine. This workstation features an *AMD Ryzen 9 5900HX* processor, 16 GB of system memory, and an *NVIDIA GeForce RTX 3060 Mobile* GPU with 6 GB of VRAM.

It is important to note that while the geometric extraction phase is computationally efficient and runs seamlessly on this consumer-grade hardware, the *SegResNet* inference mechanism requires strictly more than 6 GB of VRAM to process the segmentation of CT volumes. Consequently, the neural network forward pass could not be executed locally and relied entirely on the server infrastructure. This split-hardware approach highlights a critical threshold for future clinical deployment: while the topological analysis is highly accessible, executing the deep learning segmentation necessitates clinical workstations equipped with higher-capacity GPUs.

Chapter 5

Experiments and Discussion

5.1 Chapter Overview

This chapter presents a rigorous evaluation of the proposed framework, detailing the experimental setup, training configurations, and the comparative performance of various deep learning architectures on the dataset splits. It begins by contrasting preliminary 2D approaches with fully 3D volumetric models, demonstrating the necessity of the latter for maintaining topological continuity. The chapter provides a comprehensive quantitative and qualitative analysis of the segmentation models, ultimately highlighting the superior efficacy of specific convolutional networks, and concludes by showcasing the successful extraction and clinical relevance of the resulting morphological and topological features.

5.2 Dataset Splits

During the preliminary phases of the study, an initial data partition of 48 training, 6 validation, and 2 testing scans was utilized to establish a baseline performance metric using the *SwinUNETR* architecture. While this initial configuration intended to maximize the volume of data available for the learning phase, the constrained size of the test set limited the ability to thoroughly evaluate model generalizability across diverse anatomical variations.

To ensure a more rigorous comparative analysis and to provide a statistically robust test set, the dataset was subsequently redistributed into a standardized 70%/15%/15% configuration. This finalized allocation provided 40 scans for training, 8 for validation, and 8 for the test set. This 40/8/8 split was uniformly applied to train and evaluate the performance of all five selected architectures in the main comparative study: *3D U-Net*, *DynUNet*, *SegResNet*, *SwinUNETR*, and *UNETR*.

5.3 Training Configurations

To ensure a fair and consistent comparative analysis, all five deep learning architectures (*3D U-Net*, *DynUNet*, *SegResNet*, *SwinUNETR* and *UNETR*) were trained using an identical configuration pipeline.

The network weights were optimized using the *AdamW* optimizer, initialized with a learning rate of 2×10^{-4} and a weight decay of 1×10^{-5} . The training process was configured

for a maximum of 3000 epochs. To maintain stable convergence as the models approached local minima, a cosine annealing learning rate scheduler was applied over the total estimated stepping batches. Furthermore, training was executed using mixed precision (*bfloat16*) and a gradient accumulation of 2 steps, which optimized GPU memory utilization and accelerated the computational workflow.

The objective function applied across all models was the *Dice Focal Loss* (with $\gamma = 2.0$). This hybrid loss function was specifically selected to address the inherent class imbalance between the relatively small volumetric footprint of the AVT and the vast surrounding background tissue, simultaneously maximizing spatial overlap and penalizing highly confident misclassifications.

Given the constrained size of the training dataset (40 CTA scans), a robust, on-the-fly data augmentation strategy was implemented to prevent overfitting and force the models to learn invariant anatomical features. During the data loading pipeline, each volume was subjected to a series of randomized spatial and intensity transformations prior to patch extraction.

The spatial augmentations included random axis flipping along the sagittal, coronal, and axial planes (each with a 30% probability) and random 90-degree rotations (30% probability). To account for the radiodensity heterogeneity inherent to the multicenter dataset, intensity-based augmentations were also applied. These included random intensity shifting by an offset of 0.10 (50% probability), the injection of random Gaussian noise (30% probability), and the simulation of Gibbs ringing artifacts (30% probability). By applying these stochastic transformations, the networks were continuously exposed to novel, varied representations of the training data at every epoch.

5.4 Preliminary Experiments: 2D vs. 3D

Prior to evaluating fully volumetric architectures, preliminary experiments were conducted to assess the viability of 2D object detection and segmentation models for delineating the AVT. Specifically, a two-stage 2D YOLO pipeline was implemented to process the CTA volumes on a slice-by-slice basis. The first stage consisted of an object detection model designed to localize the aorta and output a Region of Interest (ROI) bounding box for each axial slice. The second stage utilized a segmentation model that processed this isolated ROI to generate the final vessel mask.

While the 2D pipeline demonstrated a baseline ability to identify the aortic cross-section, the decoupled, slice-by-slice approach ultimately proved inadequate for the complex topology of the anatomy. Because the detection model evaluated each slice independently without longitudinal (Z-axis) context, it frequently failed to localize the aorta in slices exhibiting poor contrast with surrounding tissues, particularly within the narrow, small-diameter branching

vessels where visibility is inherently limited. If the detection stage failed, the segmentation stage received no input, resulting in entirely missing segments and severe structural discontinuities along the length of the reconstructed vessel.

Furthermore, even when the ROI was successfully localized, the 2D segmentation predictions lacked anatomical precision. The model consistently generated overly simplified, artificially rounded, or squared-off contours. When these 2D predictions were stacked to reconstruct the 3D volume, the resulting mesh exhibited a highly voxelated, blocky appearance with severe stair-step artifacts, failing to capture the smooth, natural curvature of the vessel walls (Figure 5.1).

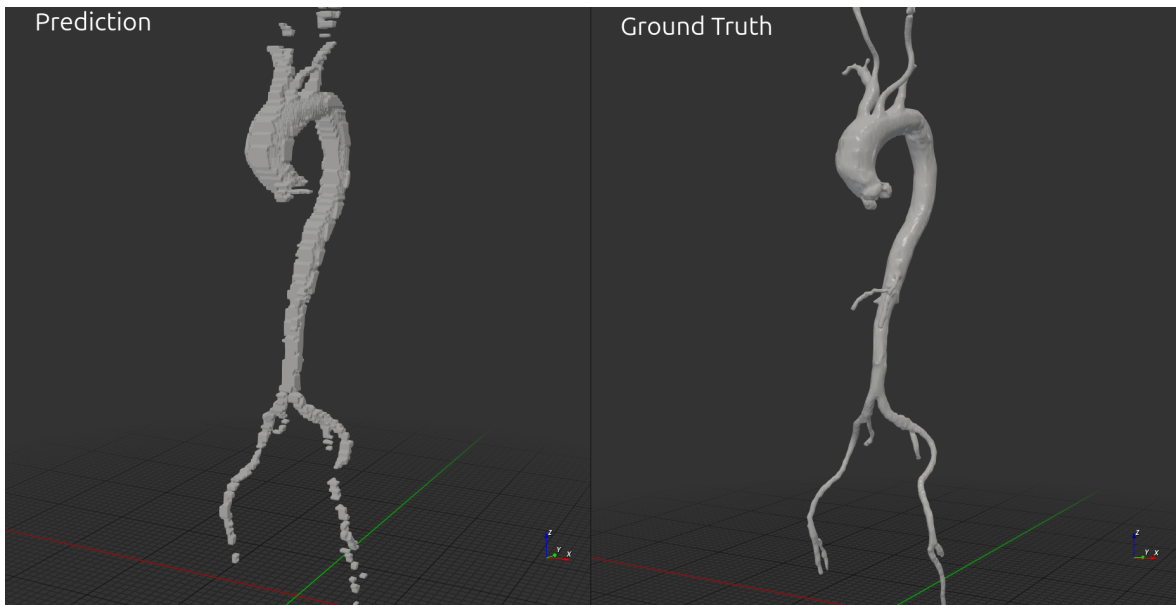


Figure 5.1: Comparison of the 3D surface reconstruction generated by the 2D YOLO slice-by-slice pipeline (left) and the Ground Truth (right). The 2D prediction exhibits severe longitudinal discontinuities due to missed slice-level detections, alongside a highly blocky, voxelated topology resulting from imprecise 2D contouring.

Given that the subsequent morphological analysis, such as centerline extraction and local curvature calculation, requires a strictly continuous and morphologically faithful surface mesh, these fragmented and artifact-heavy masks were mathematically unviable for the pipeline. These preliminary findings conclusively demonstrated that maintaining spatial continuity in the Z-axis is a strict requirement for this task, directly justifying the methodological shift toward the fully 3D architectures evaluated in the subsequent sections.

5.5 Volumetric Segmentation

The evaluation of the volumetric segmentation models relies on a comprehensive analysis of geometric accuracy and boundary precision. This section details the performance of the evaluated architectures using standard quantitative metrics and visual qualitative assessments to ensure the topological validity of the generated surface meshes.

5.5.1 Evaluation Metrics

To rigorously assess the segmentation performance, the models were evaluated using one overlap-based metric and two distance-based metrics. This combination ensures that both the global volumetric agreement and the localized boundary precision of the segmented AVT are quantified.

The primary metric utilized is the DSC, which measures the spatial overlap between the predicted segmentation mask (X) and the ground truth mask (Y), mathematically defined as:

$$DSC = \frac{2|X \cap Y|}{|X| + |Y|}$$

While DSC provides a robust measure of overall volumetric agreement, it is relatively insensitive to boundary localization errors. Therefore, distance-based metrics were incorporated to evaluate surface fidelity.

The HD95 evaluates the maximum boundary deviation while remaining robust to outlier voxels. It calculates the 95th percentile of all shortest distances from the boundary points of the prediction to the boundary points of the ground truth. A lower HD95 value indicates that the predicted boundary closely traces the true anatomical boundary.

Additionally, the Average Symmetric Surface Distance (ASSD) was utilized. ASSD calculates the average of all shortest distances between the surface points of the predicted mask and the ground truth mask, providing a generalized measure of surface-to-surface alignment. Like HD95, lower ASSD values indicate higher boundary precision.

5.5.2 Quantitative Results

To determine the most effective computational architecture for the segmentation of the AVT, a comprehensive comparative analysis was conducted. Five distinct state-of-the-art deep learning models were evaluated on the test set: *3D U-Net*, *DynUNet*, *SegResNet*, *SwinUNETR*, and *UNETR*. The aggregated average metrics for each architecture are presented in Table 5.1.

Architecture	Dice $\uparrow$	HD95 (mm) $\downarrow$	ASSD (mm) $\downarrow$
3D U-Net	0.8935	84.51	9.95
DynUNet	0.9113	59.74	7.83
SegResNet	0.9220	36.30	3.25
SwinUNETR	0.8719	111.67	12.95
UNETR	0.7335	171.60	30.87

Table 5.1: Quantitative comparison of deep learning architectures on the test set. Best results are highlighted in bold.

As the results indicate, *SegResNet* demonstrated superior performance across all evaluated metrics, achieving the highest volumetric overlap with an average DSC of 0.9220. Furthermore, *SegResNet* significantly outperformed the other models in boundary precision, recording an HD95 of 36.30mm and an ASSD of 3.25mm. *DynUNet* served as the second most effective architecture, achieving a highly competitive DSC of 0.9113, though it exhibited higher boundary errors than *SegResNet*.

Notably, while transformer-based architectures such as *SwinUNETR* and *UNETR* possess the theoretical advantage of capturing long-range global dependencies, they underperformed in this specific experimental setup. *SwinUNETR* achieved an average DSC of 0.8719, while *UNETR* struggled significantly with a DSC of 0.7335. This performance discrepancy suggests that vision transformers typically require massive datasets to learn generalized representations. Given the highly heterogeneous nature and the relatively constrained size of the AVT dataset (56 scans), the convolutional inductive biases and residual connections inherent to *SegResNet* proved vastly more efficient and robust for this specific segmentation task.

5.5.3 Qualitative Results

While quantitative metrics provide a statistical overview of performance, a visual assessment of the generated 3D surface meshes is necessary to evaluate the topological correctness of the segmentations. High overlap scores do not inherently guarantee contiguous vascular structures, which are strictly required for subsequent morphological and centerline analyses.

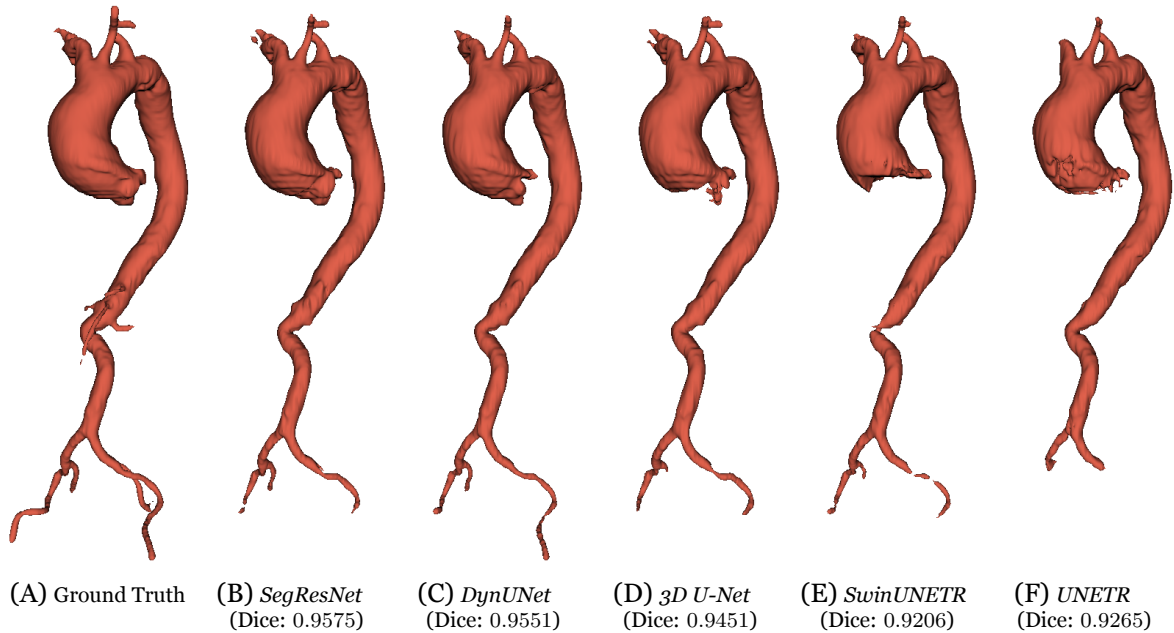


Figure 5.2: Qualitative evaluation of the second best-performing test case (D2), achieving a high average Dice score of 0.9410. The volumetric reconstructions demonstrate strong spatial agreement with the ground truth, with most architectures successfully preserving the topological continuity of the distal branches.

In high-performing instances, such as Case D2 (Figure 5.2), most architectures demonstrated

robust spatial agreement with the ground truth. *SegResNet* and *DynUNet* consistently produced smooth, continuous vascular trees, successfully delineating both the primary aortic body and the narrower supra-aortic and iliac branches without severe artifacting.

Curiously, significant performance degradation was observed in Case R4 (Figure 5.3), despite it representing a standard, non-pathological aortic geometry. The reasons for this widespread underperformance across all evaluated architectures remain unclear. *UNETR* failed almost completely, segmenting only a negligible fraction of the vessel, while *SwinUNETR* exhibited severe spatial discontinuities. Even the most robust convolutional models, *SegResNet* and *DynUNet*, struggled with this specific scan, producing visible segmentation errors and irregular, non-smooth surface boundaries. Furthermore, the distal branches, such as the iliac arteries, are partially absent in these reconstructions. This irregularities is primarily a byproduct of localized spatial discontinuities in the raw predictions. Because the MONAI `KeepLargestConnectedComponent` post-processing function is intentionally applied to eliminate random background noise and ensure a single contiguous mesh for centerline extraction, any successfully segmented but disconnected arterial branches are automatically and deliberately discarded.

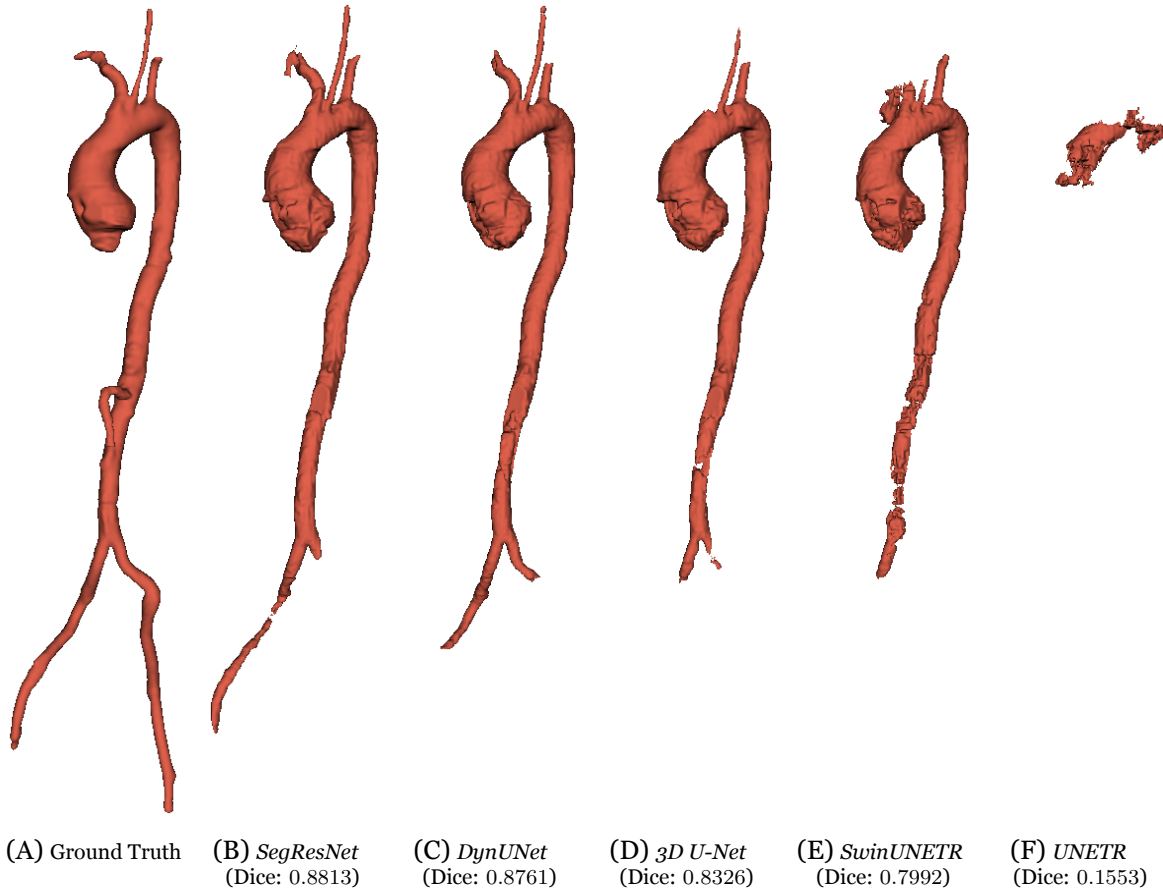


Figure 5.3: Qualitative evaluation of the worst-performing test case (R4), which recorded an average Dice score of 0.7089 across all evaluated models. Visual inspection highlights the difficulties encountered by the architectures in reconstructing the complete vascular topology, leading to prominent spatial discontinuities and fragmented branches.

In contrast to the difficulties encountered with Case R4, the models demonstrated surprisingly strong performance when processing high pathological complexity. In Case R1-AD (Figure 5.4), which features an AD with true and false lumens separated by a thin intimal flap, the convolutional models successfully reconstructed the severe morphological irregularities. While *SwinUNETR* and *UNETR* continued to perform poorly, the remaining architectures were able to accurately capture these complex internal boundaries without introducing structural deformities or surface anomalies. However, similar to Case R4, one iliac artery is missing in the final render of *SegResNet* due to minor topological breaks triggering the intentional removal of this segment during connected component analysis. Overall, *SegResNet* and *DynUNet* exhibited equivalent, high-quality visual performance in the main aortic body of this challenging scenario.

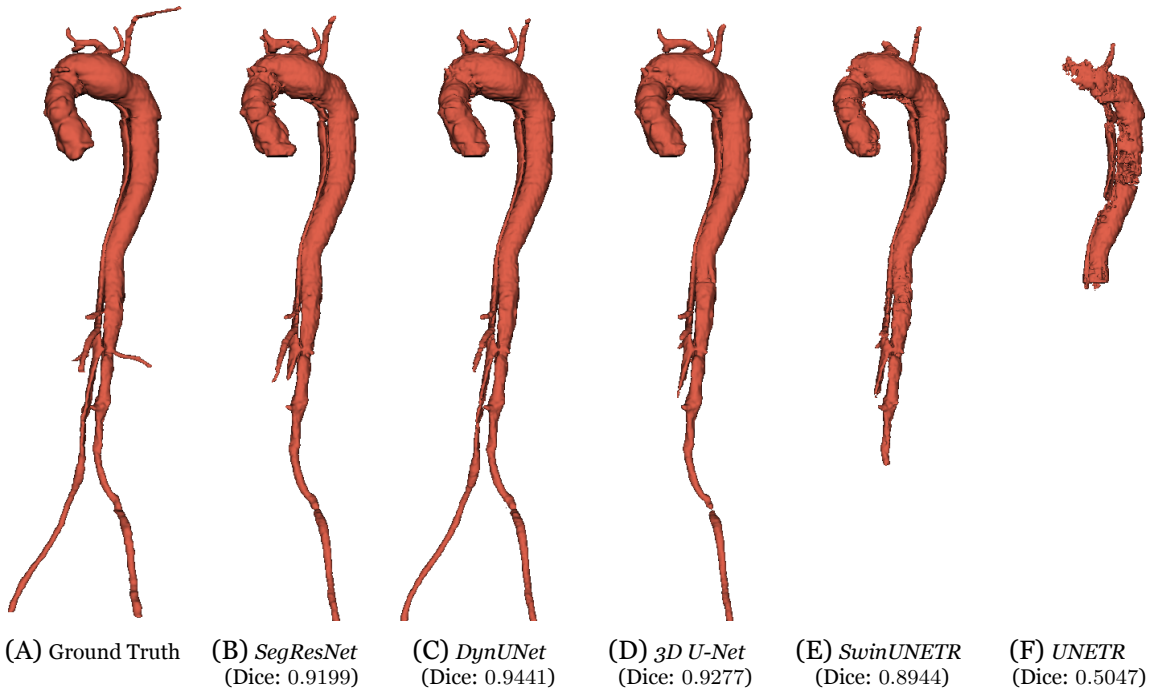


Figure 5.4: Qualitative evaluation of a highly complex pathological case exhibiting an aortic dissection (R1-AD), which achieved an average Dice score of 0.8382. The presence of the dissection flap and the resulting true and false lumens introduce significant morphological irregularities, challenging the models' ability to maintain accurate boundary delineation without structural deformities.

Despite the unexplained limitations observed in specific instances like Case R4, the overall visual consensus aligns with the quantitative data. Both *SegResNet* and *DynUNet* demonstrated an equivalent and superior ability to maintain topological integrity and smooth surface boundaries across the test set, successfully managing even complex pathologies. The removal of disconnected branches by the `KeepLargestConnectedComponent` function, while strictly necessary for the stability of the subsequent morphological centerline pipeline, exposes a critical limitation in the models' predictive continuity. Ensuring absolute spatial continuity in the raw predictions, so that valid anatomical structures are not discarded during this mandatory post-processing step, remains an ongoing challenge to be addressed in future work. Nevertheless, given its quantitative advantages, *SegResNet* confirms its suitability as the foundational segmentation model for the geometric feature extraction framework.

5.6 Morphological Analysis and Feature Extraction

Following the successful volumetric segmentation of the AVT, the resulting 3D binary masks were converted into continuous surface meshes to compute quantitative geometric biomarkers. This extraction was performed utilizing the VMTK, a specialized open-source framework designed for the advanced geometric and topological analysis of 3D vascular structures from medical imaging.

Centerlines were extracted from the aortic root to the distal endpoints of the iliac and supra-aortic branches. To ensure a comprehensive geometric profile is available for subsequent predictive modeling, the automated pipeline extracts an extensive array of morphological features at discrete intervals along the entire vascular network.

5.6.1 Spatial Coordinates and Bifurcation Dynamics

During the extraction process, two distinct sets of 3D spatial coordinates are recorded for every discrete point: the true centerline path, computed via maximum inscribed spheres, and the geometric centroid of the corresponding orthogonal cross-section. While these two coordinates are nearly identical in straight segments of the aorta, they deviate significantly near anatomical bifurcations. Approaching a branching vessel, the localized cross-section elongates and widens to encompass the origin of the new branch. This natural anatomical widening shifts the geometric centroid away from the true centerline path, while simultaneously causing the cross-sectional area and shape eccentricity to spike as seen in Figure 5.5.

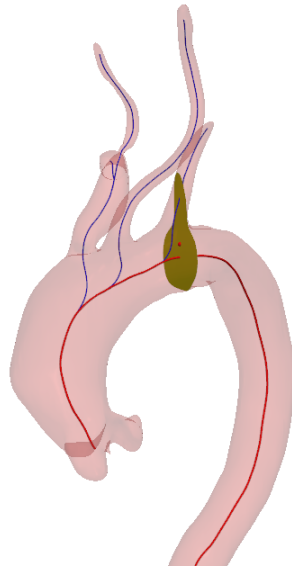


Figure 5.5: Cross-sectional elongation at an anatomical bifurcation. The orthogonal plane (yellow) widens as it intersects the emerging branch, visibly shifting its geometric centroid (red dot) away from the main centerline trajectory (red).

Capturing both spatial coordinates is critical for downstream machine learning applications. A point-wise classification model designed to detect localized anomalies, such as an AAA,

must be able to distinguish between a pathological bulge and a natural bifurcation. By analyzing the deviation between the true centerline and the cross-section centroid, the predictive model can contextualize extreme fluctuations in area and shape, correctly interpreting them as the geometric signature of a branching event rather than a false-positive anomaly.

5.6.2 Cross-Sectional Morphometry

To accurately assess aortic health, the automated pipeline evaluates the cross-sectional size and the regularity of the vessel wall. These features are calculated strictly orthogonal to the vessel's local trajectory, preventing the overestimation of dimensions in highly angulated regions. The true internal diameter of the vessel is calculated via the maximum inscribed sphere (Figure 5.6), which ensures a strict orthogonal measurement that remains unaffected by the local angulation of the aorta.

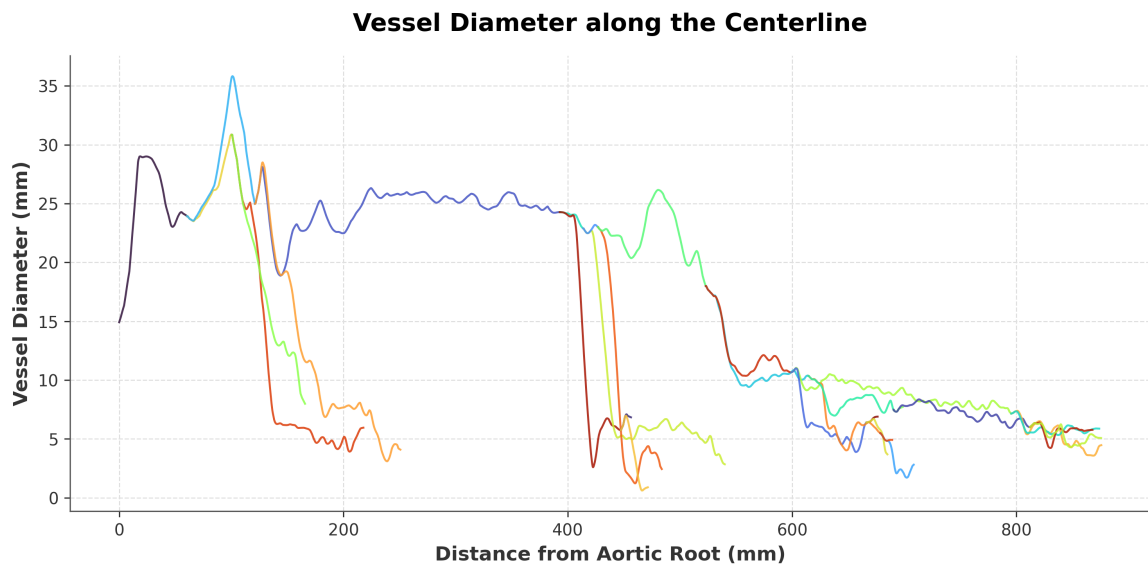


Figure 5.6: Vessel diameter measurement (mm) plotted against the distance from the aortic root (mm). Tracking the diameter continuously along the longitudinal axis allows for the precise spatial localization of aneurysmal dilations or stenotic narrowing across all branching vessels.

The cross-sectional slice itself is further quantified by its total luminal area (Figure 5.7), as well as its minimum and maximum spatial dimensions, which represent the minor and major axes of the slice.

The ratio between these minimum and maximum dimensions yields a shape index (Figure 5.8). This metric directly translates to the eccentricity of the vessel lumen. A perfectly circular cross-section yields a shape index of 1.0, while a flattened, elliptical, or asymmetrically remodeled aneurysmal wall yields a significantly lower value.

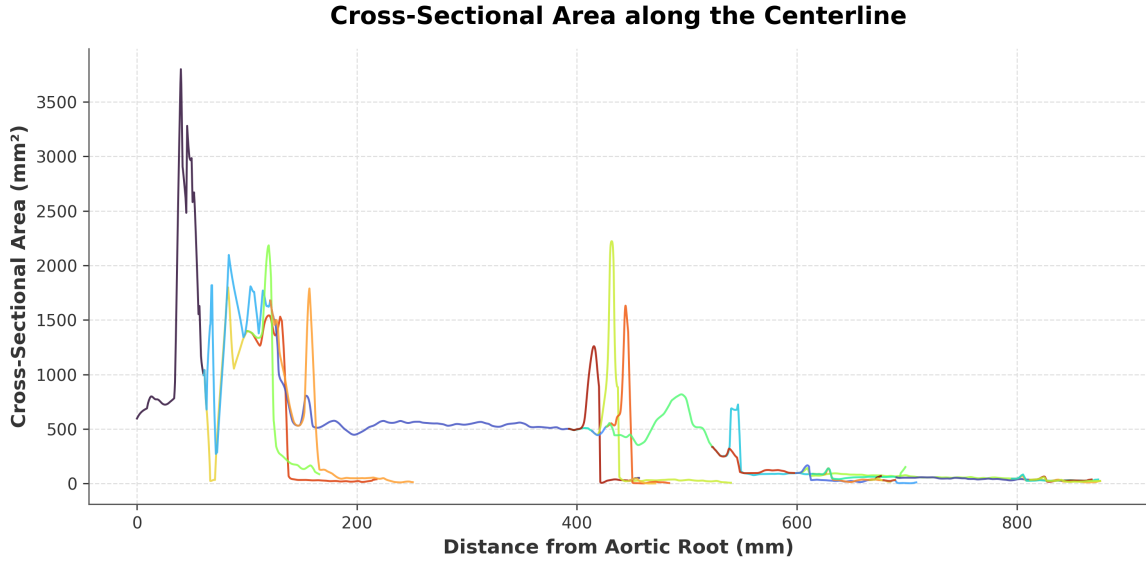


Figure 5.7: Cross-sectional area (mm^2) along the centerline. Area measurements provide a highly sensitive volumetric indicator of vessel expansion, capturing asymmetric dilations that standard unidimensional diameter measurements may underestimate.

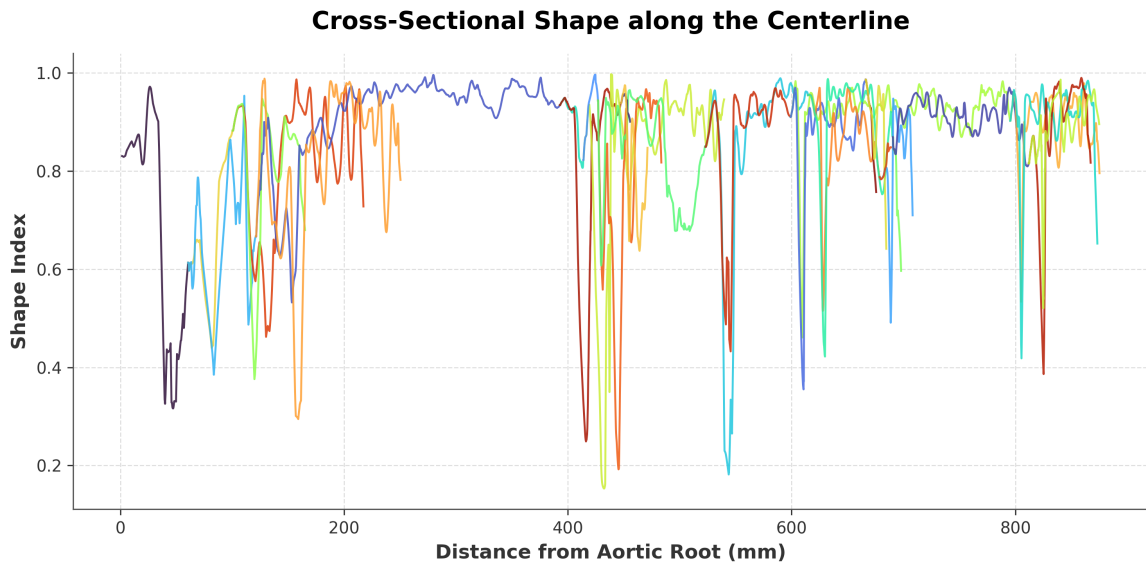


Figure 5.8: Cross-sectional shape index plotted against the centerline distance. This metric quantifies the eccentricity of the vessel lumen, where values closer to 1.0 indicate a perfect circle, and lower values indicate elliptical or irregular wall flattening.

5.6.3 3D Path Dynamics and Topology

Beyond cross-sectional morphometry, the 3D topological trajectory of the aorta heavily influences hemodynamic flow. To quantify this complex geometry, the framework computes local curvature, torsion, and tortuosity.

Curvature (Figure 5.9) measures the degree to which the vessel deviates from a straight line at a specific spatial coordinate. Elevated localized curvature values often correlate with areas subjected to abnormal hemodynamic forces.

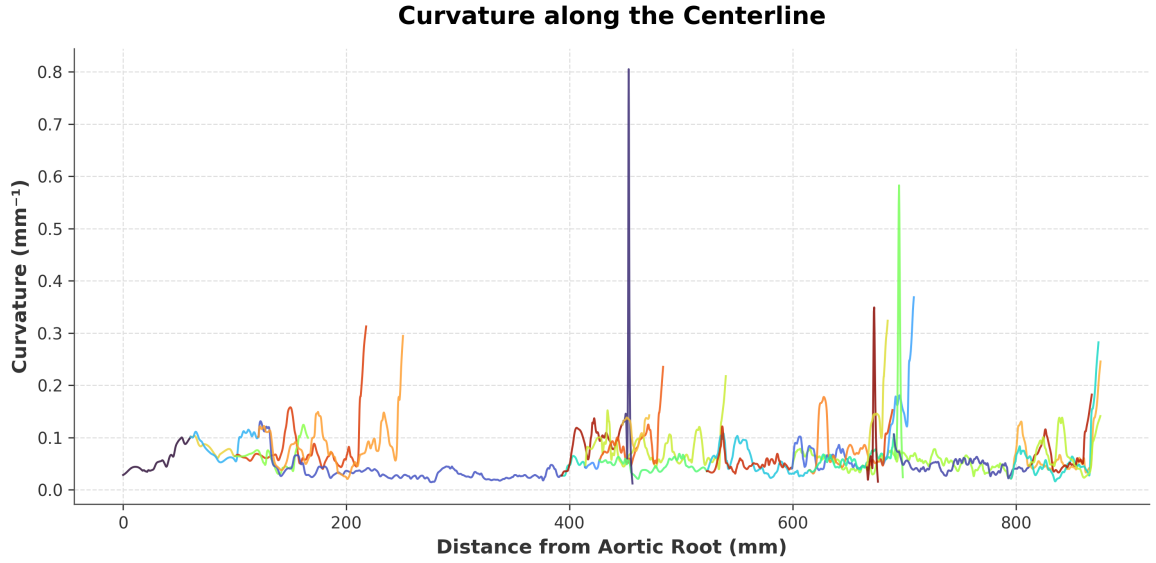


Figure 5.9: Local curvature (mm^{-1}) plotted along the centerline. Curvature measures the spatial deviation from a straight trajectory. High-frequency spikes indicate sharp bends or topological irregularities, often corresponding to anatomical bifurcations or regions of disturbed hemodynamic flow.

Similarly, torsion (Figure 5.10) quantifies the 3D twisting behavior of the vessel. To fully formalize this 3D trajectory mathematically, the framework extracts the Frenet-Serret frames at every discrete coordinate. The Frenet frame defines a moving local coordinate system consisting of three orthogonal vectors:

- **Tangent Vector:** Indicates the forward, longitudinal direction of the vessel path.
- **Normal Vector:** Points in the direction of the path's curvature or bending.
- **Binormal Vector:** The cross product of the tangent and normal vectors, describing the torsion or spatial twisting of the path.

Together, these vectors provide a complete mathematical descriptor of the vessel's spatial orientation at any given point.

To maintain the structural integrity of this vast dataset, topological connectivity variables, known as edge arrays, are also exported. These edge variables define the precise mathematical graph structure of the vascular tree, establishing exactly which discrete spatial points connect to form continuous branches, and explicitly marking the indices where bifurcations split.

Finally, the framework evaluates vessel tortuosity (Figure 5.11). Rather than computing localized segment values, the tortuosity ratio is calculated cumulatively from the aortic root to each successive point along the centerline, dividing the total accumulated path length by the direct Euclidean distance from the origin. Starting at a baseline value of 1.0 at the root, the

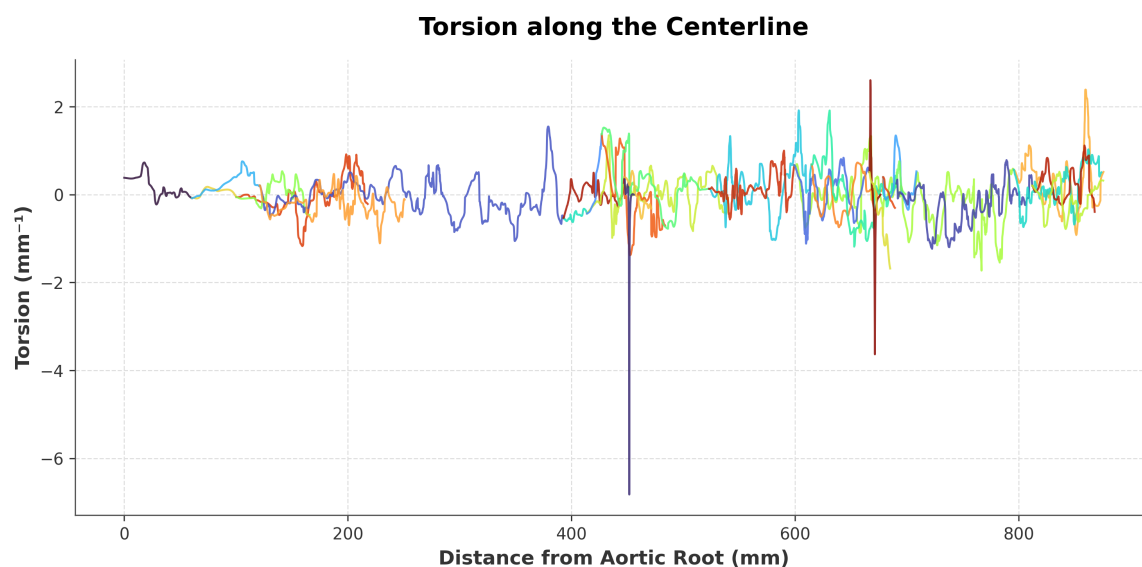


Figure 5.10: Torsion (mm^{-1}) plotted along the centerline. Torsion quantifies the extent to which the vessel twists out of a 2D plane into a 3D spiral configuration, a metric highly relevant for assessing the morphological complexity of aortic dissections.

metric steadily increases along the descending aorta, reflecting the cumulative spatial deviation and bending of the global vessel trajectory. Evaluating this overall trajectory is clinically relevant, as it quantifies the general geometric deformation of the aorta across its entire longitudinal extension, providing a macroscopic index of vascular elongation. By systematically isolating and exporting these geometric, spatial, and topological features, the computational framework provides an exhaustive, machine-learning-ready dataset for advanced cardiovascular analysis.

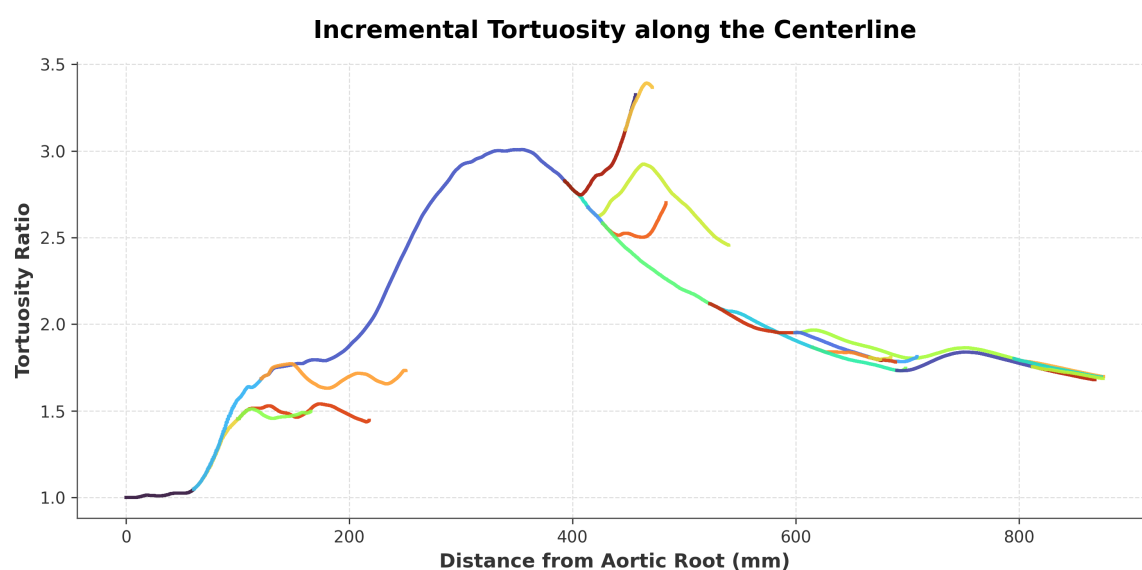


Figure 5.11: Incremental tortuosity ratio plotted along the distance from the aortic root. This metric tracks the coiling severity of the vessel continuously, providing a quantitative assessment of longitudinal vessel deformation.

5.7 Discussion

The development of the proposed computational framework demonstrates the feasibility of bridging raw, multi-center radiological data with advanced geometric analysis to automate the extraction of cardiovascular biomarkers. By integrating deep learning-based volumetric segmentation with mathematical surface modeling, the pipeline successfully translates discrete voxel data into a continuous, machine-learning-ready dataset consisting of 28 morphological and topological features.

5.7.1 Segmentation Performance and Architectural Trade-offs

The comparative analysis of volumetric segmentation models revealed a distinct performance disparity between convolutional architectures and vision transformers. *SegResNet* achieved the highest quantitative results across all metrics, securing an average DSC of 0.9220, while transformer-based models like *UNETR* and *SwinUNETR* yielded sub-optimal performance.

This discrepancy highlights a well-documented characteristic of transformer architectures: their reliance on massive datasets. Vision transformers utilize self-attention mechanisms to model global context, lacking the inherent inductive biases, such as translation invariance and localized spatial correlation, that CNNs possess. In the context of the AVT dataset, which contains only 56 highly heterogeneous CTA scans, the dataset size was likely insufficient for the transformer models to learn generalized global representations effectively. Conversely, the hierarchical convolutions and residual connections of *SegResNet* and *DynUNet* proved highly efficient at extracting robust features from a constrained amount of data, accurately delineating complex structures like the dissection flap in AD cases.

Furthermore, the necessity of rigorous preprocessing cannot be overstated. The multi-center nature of the dataset introduced severe variations in spatial resolution and radiodensity baselines. The implementation of isotropic resampling to $1.0mm$ and global Z-score normalization was critical in neutralizing scanner-induced artifacts, ensuring that the architectures evaluated the anatomy at a consistent physical scale.

5.7.2 Topological Continuity and Pipeline Limitations

Despite the strong overlap metrics achieved by the convolutional models, the qualitative analysis exposed a critical vulnerability regarding spatial continuity. In tubular structures like the aorta, maintaining absolute topological integrity is paramount.

The pipeline relies on the MONAI `KeepLargestConnectedComponent` function as an intentional, mandatory post-processing step to eliminate background noise and isolate a single, contiguous vascular mesh for the VMTK algorithms. However, this strict requirement exposes a limitation in the models' predictive continuity. If a model fails to segment even a

single voxel connecting a distal branch (such as an iliac artery) to the main aortic body, that entire branch is deliberately discarded by the post-processing filter. This phenomenon explains the unexpected visual absences of distal branches in otherwise healthy scans, such as Case R4. Addressing this limitation remains a primary objective for future development.

5.7.3 Clinical Applicability of Geometric Feature Extraction

The geometric feature extraction phase successfully moves the clinical evaluation of the aorta beyond simplified 2D approximations. Traditional clinical guidelines heavily rely on maximum diameter thresholds to assess the rupture risk of an AAA. However, diameter alone fails to capture the asymmetric remodeling and spatial elongation that characterize progressive vascular diseases.

By calculating features continuously along the centerline, the framework provides an exhaustive geometric profile. The extraction of cross-sectional area and shape eccentricity provides a localized assessment of asymmetric dilations. Concurrently, calculating 3D path dynamics, including local curvature, torsion, and the orthogonal Frenet-Serret vectors, maps the tortuous trajectory of the vessel. These specific parameters are directly correlated with abnormal hemodynamic wall shear stress. By systematically exporting these metrics into a structured tabular format, the pipeline provides the foundational data required to train point-wise classification models capable of distinguishing healthy bifurcations from pathological anomalies.

5.7.4 Computational Efficiency and Pipeline Automation

For an automated diagnostic framework to be viable in a clinical setting, it must operate within practical hardware and temporal constraints. The transition from 2D slice-by-slice models to fully 3D architectures exponentially increases the computational RAM use.

Processing high resolution volumetric scans requires significant GPU memory. During the sliding window inference phase, executed on an *NVIDIA A40* GPU, the convolutional models required a peak VRAM ranging from 12.67 GB for standard volumes up to 32.04 GB for the highest resolution scans. Despite this memory variance, the network proved highly time efficient, completing the segmentation in an average of 56.7 seconds per volume. Furthermore, the system RAM use remained remarkably lightweight, consistently operating under 5.0 GB. While transformer based models generally demand higher computational overhead, *SegResNet* remained strictly within an acceptable envelope for modern clinical workstations.

Once the segmentation is complete, the VMTK topological extraction and geometric analysis rely on a semi-automated workflow. The system pauses for manual intervention only during the initial spatial ray casting to define the aortic root, after which it resumes autonomous execution. Following this initialization, the primary centerline extraction executes swiftly,

generally concluding in under 2 to 5 minutes. However, performance profiling of the entire test set reveals that the subsequent computation of orthogonal cross-sections represents the pipeline’s primary computational bottleneck. For particularly dense or complex vascular networks, this step can extend processing times significantly (up to 25 minutes) and occasionally induces memory segmentation faults on extreme resolution topologies.

For the majority of successfully processed cases, the complete end-to-end diagnostic pipeline, from raw NIfTI image to fully extracted geometric features, concludes in approximately 3.5 to 13.5 minutes. Despite the structural complexities of calculating discrete orthogonal planes along elongated branches, this semi-automated approach drastically reduces the overall time required for morphological assessment compared to manual slice-by-slice contouring, offering a highly reproducible and time efficient workflow for longitudinal patient monitoring and advanced cardiovascular research.

5.7.5 Weaknesses and Further Work

Despite the robust performance of the proposed framework, several critical weaknesses remain, primarily stemming from the sequential and compounding nature of the automated pipeline. The foundational limitation is that the volumetric segmentation is not entirely flawless. Minor voxel misclassifications can have a disproportionate impact due to the mandatory `KeepLargestConnectedComponent` post-processing function. If the segmentation model fails to predict a continuous connection to a distal branch by even a single voxel, the algorithm completely discards the disconnected branch. This phenomenon occurs frequently in the narrower iliac arteries, leading to the artificial absence of valid anatomical structures as seen in Figures 5.3 and 5.4.

Furthermore, even when an artery maintains macro-level connectivity, localized segmentation inaccuracies can drop one or two voxels, creating artificially “squeezed” spots with extremely low diameters. The mathematical algorithms within the VMTK frequently fail to traverse these tight topological bottlenecks. As a result, the centerline extraction may terminate prematurely, failing to map the full length of the branch even if the complete branch segment is present in the volumetric mask. This specific failure mode is also highly prevalent in the iliac arteries.

These spatial and topological anomalies create an additional point of failure by triggering a cascade of subsequent errors. A premature centerline termination directly impacts the next geometric phase: the computation of orthogonal cross-sections. In certain cases, such as test Case D1, the lack of centerline continuity prevents the cross-sectional algorithms from executing in that branch, throwing a fatal error that breaks the entire pipeline for that specific CT scan. Consequently, the final exported numerical table is left incomplete, featuring fewer columns because the framework could not compute the respective cross-sectional geometric features.

However, the severity of this error cascade is variable. Because the framework processes the vascular tree as a complex graph, an early morphological error does not universally corrupt the entire scan. In many instances, the segmentation or centerline extraction might fail in a single iliac artery, yet the pipeline will still successfully compute and export the complete suite of features for the remainder of the vascular anatomy. Addressing these strict connectivity dependencies and improving the robustness of the centerline extraction against minor segmentation flaws remain primary objectives for future iterations of this work.

Chapter 6

Conclusions and Future Work

6.1 Conclusions

This dissertation developed and evaluated an end-to-end automated computational framework for the volumetric segmentation and advanced geometric analysis of the AVT from CTA scans. By systematically comparing state-of-the-art deep learning architectures, the research demonstrated that convolutional models, specifically *SegResNet*, significantly outperformed transformer-based alternatives in accurately delineating the complex and tortuous boundaries of the aorta within a constrained multi-center dataset.

Beyond semantic segmentation, the proposed pipeline bridged the gap between discrete voxel representations and continuous mathematical modeling by integrating automated surface mesh generation, topological mapping, and centerline extraction. The framework successfully automated the computation of a comprehensive suite of morphological biomarkers, including cross-sectional diameter, shape eccentricity, local curvature, torsion, and overall tortuosity. By outputting these localized features as a structured numerical dataset, the system provides a far more robust and multidimensional characterization of aortic pathologies than traditional unidimensional clinical metrics, establishing a highly reproducible foundation for computer-aided diagnosis and longitudinal disease monitoring.

6.2 Future Work

While the proposed framework successfully automates the extraction of geometric features, several areas for improvement and expansion have been identified for future development:

- **3D Bounding Box Anatomical Isolation:** Currently, the preprocessing pipeline utilizes a foreground cropping method that, while reducing background space, still leaves extraneous regions and scanner hardware within the volumetric bounds. Future iterations should train and integrate a dedicated 3D bounding box object detection model to precisely isolate the patient’s true physical anatomy. By ensuring the subsequent segmentation network only evaluates the relevant anatomical region, this targeted isolation will significantly reduce the number of voxels processed, thereby increasing computational speed and improving segmentation accuracy by eliminating irrelevant background artifacts.
- **Dataset Expansion with Complex Pathologies:** The generalization capabilities of the segmentation models, particularly regarding topological continuity in the distal branches and highly abnormal structures, were limited by the relatively small size of

the 56-scan multi-center dataset. Future work must involve training the segmentation architectures on a substantially larger cohort of CT scans. Incorporating a higher volume of complex pathological cases, specifically diverse AAs and severe ADs, will heavily enhance the network's structural robustness and its ability to seamlessly generalize across extreme vascular deformations.

- **Predictive Machine Learning Modeling:** The final output of the current pipeline is a rich, structured dataset containing dozens of localized geometric and topological features. The natural progression of this research is to utilize this numerical data to train downstream machine learning prediction models. Developing point-wise classification or regression models will enable the system to autonomously distinguish healthy natural bifurcations from pathological anomalies, effectively predicting localized regions of high rupture risk and fully realizing the diagnostic potential of the extracted morphological biomarkers.

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